Concept 1 | Coordinate Geometry Basics

English Student Edition • Focused formulas and guided practice

Formula opener

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint

$$M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$$

Gradient

$$m = \frac{y_2-y_1}{x_2-x_1}$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Coordinate Geometry Basics**

Q001 Written

A (-4), B (8): distance AB

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 12.
3. Final answer: 12.

Final answer

12

Q002 Written

A (-5), B (7): distance AB

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 12.
3. Final answer: 12.

Final answer

12

Worked Solutions **Coordinate Geometry Basics**

Q003 Written

For two points A (-2) and B (6), find the distance AB.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 8.
3. Final answer: 8.

Final answer

8

Q004 Written

For two points A (4) and B (10), find the distance AB.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 6.
3. Final answer: 6.

Final answer

6

Worked Solutions **Coordinate Geometry Basics**

Q005 MCQ

Find a (-4), B (8): distance AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 12.

Correct option: C

Final answer

12

Q006 MCQ

Find a (-5), B (7): distance AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 12.

Correct option: C

Final answer

12

Worked Solutions **Coordinate Geometry Basics**

Q007 MCQ

For two points A (-2) and B (6), find the distance AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 8.

Correct option: C

Final answer

8

Q008 MCQ

For two points A (4) and B (10), find the distance AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 6.

Correct option: C

Final answer

6

Worked Solutions **Coordinate Geometry Basics**

Q009 **Written**

A (-4), B (8): internal point C in ratio 2:1

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 4.
3. Final answer: 4.

Final answer

4

Q010 **Written**

A (-5), B (7): internal point C in ratio 2:1

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 3.
3. Final answer: 3.

Final answer

3

Worked Solutions **Coordinate Geometry Basics**

Q011 Written

For two points A (-2) and B (6), find the coordinate of point C dividing segment AB internally in the ratio 3:1.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 4.
3. Final answer: 4.

Final answer

4

Q012 Written

For two points A (4) and B (10), find the coordinate of point C dividing segment AB internally in the ratio 2:1.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 8.
3. Final answer: 8.

Final answer

8

Worked Solutions **Coordinate Geometry Basics**

Q013 MCQ

Find a (-4), B (8): internal point C in ratio 2:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 4.

Correct option: B

Final answer

4

Q014 MCQ

Find a (-5), B (7): internal point C in ratio 2:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 3.

Correct option: C

Final answer

3

Worked Solutions   **Coordinate Geometry Basics**

Q015 MCQ

For two points A (-2) and B (6), find the coordinate of point C dividing segment AB internally in the ratio 3:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 4.

Correct option: B

Final answer

4

Q016 MCQ

For two points A (4) and B (10), find the coordinate of point C dividing segment AB internally in the ratio 2:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 8.

Correct option: B

Final answer

8

Worked Solutions **Coordinate Geometry Basics**

Q017 Written

A (-4), B (8): external point D in ratio 2:3

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives -28.
3. Final answer: -28.

Final answer

-28

Q018 Written

A (-5), B (7): external point D in ratio 2:5

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives -13.
3. Final answer: -13.

Final answer

-13

Worked Solutions **Coordinate Geometry Basics**

Q019 Written

For two points A (-2) and B (6), find the coordinate of point D dividing segment AB externally in the ratio 5:3.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 18.
3. Final answer: 18.

Final answer

18

Q020 Written

For two points A (4) and B (10), find the coordinate of point D dividing segment AB externally in the ratio 1:3.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 1.
3. Final answer: 1.

Final answer

1

Worked Solutions **Coordinate Geometry Basics**

Q021 MCQ

Find a (-4), B (8): external point D in ratio 2:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -28.

Correct option: A

Final answer

-28

Q022 MCQ

Find a (-5), B (7): external point D in ratio 2:5:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -13.

Correct option: B

Final answer

-13

Worked Solutions **Coordinate Geometry Basics**

Q023 MCQ

For two points A (-2) and B (6), find the coordinate of point D dividing segment AB externally in the ratio 5:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 18.

Correct option: A

Final answer

18

Q024 MCQ

For two points A (4) and B (10), find the coordinate of point D dividing segment AB externally in the ratio 1:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 1.

Correct option: B

Final answer

1

Worked Solutions **Coordinate Geometry Basics**

Q025 Written

A (-4), B (8): midpoint M

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 2.
3. Final answer: 2.

Final answer

2

Q026 Written

A (-5), B (7): midpoint M

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 1.
3. Final answer: 1.

Final answer

1

Worked Solutions **Coordinate Geometry Basics**

Q027 **Written**

For two points A (-2) and B (6), find the coordinate of midpoint M of segment AB.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 2.
3. Final answer: 2.

Final answer

2

Q028 **Written**

For two points A (4) and B (10), find the coordinate of midpoint M of segment AB.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 7.
3. Final answer: 7.

Final answer

7

Worked Solutions **Coordinate Geometry Basics**

Q029 MCQ

Find a (-4), B (8): midpoint M:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 2.

Correct option: C

Final answer

2

Q030 MCQ

Find a (-5), B (7): midpoint M:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 1.

Correct option: A

Final answer

1

Worked Solutions **Coordinate Geometry Basics**

Q031 MCQ

For two points A (-2) and B (6), find the coordinate of midpoint M of segment AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 2.

Correct option: D

Final answer

2

Q032 MCQ

For two points A (4) and B (10), find the coordinate of midpoint M of segment AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 7.

Correct option: A

Final answer

7

Worked Solutions

Coordinate Geometry Basics

Q033 Challenge

Pods A (-8) and B (-3): fibre-optic tether length

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is 5.

Final answer

5

Worked Solutions **Coordinate Geometry Basics**

Q034 Challenge

Pods A (-8), B (-3): internal sensor point in ratio 1:4

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is -7.

Final answer

-7

Q035 Challenge

Pods A (-8), B (-3): external outpost in ratio 3:4

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is -23.

Final answer

-23

Q036 Challenge

Pods A (-8), B (-3): midpoint supply hub

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $-11/2$.

Final answer

$-11/2$

Worked Solutions **Coordinate Geometry Basics**

Q037 Challenge

Find pods A (-8) and B (-3): fibre-optic tether length:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: A

Final answer

5

Worked Solutions **Coordinate Geometry Basics**

Q038 Challenge

Find pods A (-8), B (-3): internal sensor point in ratio 1:4:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -7.

Correct option: C

Final answer

-7

Worked Solutions **Coordinate Geometry Basics**

Q039 Challenge

Find pods A (-8), B (-3): external outpost in ratio 3:4:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -23.

Correct option: B

Final answer

-23

Worked Solutions **Coordinate Geometry Basics**

Q040 Challenge

Find pods A (-8), B (-3): midpoint supply hub:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $-11/2$.

Correct option: C

Final answer

$-11/2$

Worked Solutions **Coordinate Geometry Basics**

Q041 **Written**

Point a has coordinates (2,3): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is I.

Final answer

I

Q042 **Written**

Point b has coordinates (2,-3): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is IV.

Final answer

IV

Q043 Written

Point c has coordinates (-2,3): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is II.

Final answer

II

Q044 Written

Point d has coordinates (-2,-3): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is III.

Final answer

III

Worked Solutions **Coordinate Geometry Basics**

Q045 **Written**

Point a has coordinates (-4,2): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is II.

Final answer

II

Q046 **Written**

Point b has coordinates (1,2): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is I.

Final answer

I

Worked Solutions **Coordinate Geometry Basics**

Q047 Written

Point c has coordinates (3,-2): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is IV.

Final answer

IV

Q048 Written

Point d has coordinates (-3,-1): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is III.

Final answer

III

Worked Solutions **Coordinate Geometry Basics**

Q049 **Written**

Point a has coordinates (-3,1): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is II.

Final answer

II

Q050 **Written**

Point b has coordinates (2,-8): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is IV.

Final answer

IV

Worked Solutions **Coordinate Geometry Basics**

Q051 Written

Point c has coordinates (4,4): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is I.

Final answer

I

Q052 Written

Point d has coordinates (-6,-4): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is III.

Final answer

III

Worked Solutions **Coordinate Geometry Basics**

Q053 Written

Point e has coordinates (-2,6): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is II.

Final answer

II

Q054 Written

Point f has coordinates (3,-3): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is IV.

Final answer

IV

Worked Solutions **Coordinate Geometry Basics**

Q055 Written

Point g has coordinates (-5,-7): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is III.

Final answer

III

Q056 Written

Point h has coordinates (1,4): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is I.

Final answer

I

Worked Solutions **Coordinate Geometry Basics**

Q057 MCQ

Point a has coordinates (2,3): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is I.

Correct option: C

Final answer

I

Q058 MCQ

Point b has coordinates (2,-3): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is IV.

Correct option: B

Final answer

IV

Worked Solutions **Coordinate Geometry Basics**

Q059 MCQ

Point c has coordinates (-2,3): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is II.

Correct option: A

Final answer

II

Q060 MCQ

Point d has coordinates (-2,-3): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is III.

Correct option: C

Final answer

III

Worked Solutions **Coordinate Geometry Basics**

Q061 MCQ

Point a has coordinates (-4,2): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is II.

Correct option: A

Final answer

II

Q062 MCQ

Point b has coordinates (1,2): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is I.

Correct option: C

Final answer

I

Worked Solutions **Coordinate Geometry Basics**

Q063 MCQ

Point c has coordinates (3,-2): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is IV.

Correct option: B

Final answer

IV

Q064 MCQ

Point d has coordinates (-3,-1): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is III.

Correct option: A

Final answer

III

Worked Solutions **Coordinate Geometry Basics**

Q065 MCQ

Point a has coordinates (-3,1): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is II.

Correct option: B

Final answer

II

Q066 MCQ

Point b has coordinates (2,-8): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is IV.

Correct option: A

Final answer

IV

Worked Solutions **Coordinate Geometry Basics**

Q067 MCQ

Point c has coordinates (4,4): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is I.

Correct option: A

Final answer

I

Q068 MCQ

Point d has coordinates (-6,-4): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is III.

Correct option: C

Final answer

III

Worked Solutions **Coordinate Geometry Basics**

Q069 MCQ

Point e has coordinates (-2,6): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is II.

Correct option: B

Final answer

II

Q070 MCQ

Point f has coordinates (3,-3): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is IV.

Correct option: C

Final answer

IV

Worked Solutions **Coordinate Geometry Basics**

Q071 MCQ

Point g has coordinates (-5,-7): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is III.

Correct option: C

Final answer

III

Q072 MCQ

Point h has coordinates (1,4): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is I.

Correct option: D

Final answer

I

Worked Solutions **Coordinate Geometry Basics**

Q073 Written

For point A $(-3,\ 2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-3,\ -2)$.

Final answer

$(-3,\ -2)$

Q074 Written

For point A $(-1,\ 2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-1,\ -2)$.

Final answer

$(-1,\ -2)$

Worked Solutions **Coordinate Geometry Basics**

Q075 **Written**

For point A $(-4,\ -2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-4,\ 2)$.

Final answer

$(-4,\ 2)$

Q076 **Written**

For point A $(1,\ -2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(1,\ 2)$.

Final answer

$(1,\ 2)$

Worked Solutions **Coordinate Geometry Basics**

Q077 Written

For point A $(-3,\ 4)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-3,\ -4)$.

Final answer

$(-3,\ -4)$

Q078 MCQ

For point A $(-3,\ 2)$, find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-3,\ -2)$.

Correct option: C

Final answer

$(-3,\ -2)$

Worked Solutions **Coordinate Geometry Basics**

Q079 MCQ

For point A $(-1,\ \ 2)$, find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-1,\ \ -2)$.

Correct option: D

Final answer

$(-1,\ \ -2)$

Q080 MCQ

For point A $(-4,\ \ -2)$, find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-4,\ \ 2)$.

Correct option: C

Final answer

$(-4,\ \ 2)$

Worked Solutions **Coordinate Geometry Basics**

Q081 *MCQ*

For point A $(1,\quad -2)$, find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(1,\quad 2)$.

Correct option: D

Final answer

$(1,\quad 2)$

Q082 *MCQ*

For point A $(-3,\quad 4)$, find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-3,\quad -4)$.

Correct option: C

Final answer

$(-3,\quad -4)$

Worked Solutions **Coordinate Geometry Basics**

Q083 Written

For point A $(-3,\ 2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3,\ 2)$.

Final answer

$(3,\ 2)$

Q084 Written

For point A $(-1,\ 2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(1,\ 2)$.

Final answer

$(1,\ 2)$

Q085 Written

For point A $(-4,\ \ -2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(4,\ \ -2)$.

Final answer

$(4,\ \ -2)$

Q086 Written

For point A $(1,\ \ -2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-1,\ \ -2)$.

Final answer

$(-1,\ \ -2)$

Worked Solutions   **Coordinate Geometry Basics**

Q087 **Written**

For point A $(-3,\ \ 4)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3,\ \ 4)$.

Final answer

$(3,\ \ 4)$

Q088 **MCQ**

For point A $(-3,\ \ 2)$, find the coordinates of point Q symmetric with respect to the y-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3,\ \ 2)$.

Correct option: C

Final answer

$(3,\ \ 2)$

Worked Solutions **Coordinate Geometry Basics**

Q089 *MCQ*

For point A $(-1, \;\; 2)$, find the coordinates of point Q symmetric with respect to the y-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(1, \;\; 2)$.

Correct option: C

Final answer

$(1, \;\; 2)$

Q090 *MCQ*

For point A $(-4, \;\; -2)$, find the coordinates of point Q symmetric with respect to the y-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4, \;\; -2)$.

Correct option: A

Final answer

$(4, \;\; -2)$

Worked Solutions **Coordinate Geometry Basics**

Q091 MCQ

For point A $(1, \ -2)$, find the coordinates of point Q symmetric with respect to the y-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-1, \ -2)$.

Correct option: A

Final answer

$(-1, \ -2)$

Q092 MCQ

For point A $(-3, \ 4)$, find the coordinates of point Q symmetric with respect to the y-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, \ 4)$.

Correct option: A

Final answer

$(3, \ 4)$

Worked Solutions **Coordinate Geometry Basics**

Q093 Written

For point A $(-3,\ \ 2)$, find the coordinates of point R symmetric with respect to the origin.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3,\ \ -2)$.

Final answer

$(3,\ \ -2)$

Q094 Written

For point A $(-1,\ \ 2)$, find the coordinates of point R symmetric with respect to the origin.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(1,\ \ -2)$.

Final answer

$(1,\ \ -2)$

Worked Solutions **Coordinate Geometry Basics**

Q095 **Written**

For point A $(-4,\ -2)$, find the coordinates of point R symmetric with respect to the origin.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(4,\ 2)$.

Final answer

$(4,\ 2)$

Q096 **Written**

For point A $(1,\ -2)$, find the coordinates of point R symmetric with respect to the origin.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-1,\ 2)$.

Final answer

$(-1,\ 2)$

Worked Solutions **Coordinate Geometry Basics**

Q097 Written

For point A $(-3,\ 4)$, find the coordinates of point R symmetric with respect to the origin.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3,\ -4)$.

Final answer

$(3,\ -4)$

Q098 MCQ

For point A $(-3,\ 2)$, find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3,\ -2)$.

Correct option: B

Final answer

$(3,\ -2)$

Worked Solutions **Coordinate Geometry Basics**

Q099 MCQ

For point A $(-1,\ \ 2)$, find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(1,\ \ -2)$.

Correct option: B

Final answer

$(1,\ \ -2)$

Q100 MCQ

For point A $(-4,\ \ -2)$, find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4,\ \ 2)$.

Correct option: B

Final answer

$(4,\ \ 2)$

Worked Solutions **Coordinate Geometry Basics**

Q101 MCQ

For point A $(1, \ -2)$, find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-1, \ 2)$.

Correct option: A

Final answer

$(-1, \ 2)$

Q102 MCQ

For point A $(-3, \ 4)$, find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, \ -4)$.

Correct option: D

Final answer

$(3, \ -4)$

Worked Solutions **Coordinate Geometry Basics**

Q103 **Challenge**

A robotic welding arm is programmed at point A $(-3,\ 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(-3,\ -4)$.

Final answer

$(-3,\ -4)$

Worked Solutions **Coordinate Geometry Basics**

Q104 Challenge

A robotic welding arm is programmed at point A $(-3,\ 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(3,\ 4)$.

Final answer

$(3,\ 4)$

Worked Solutions **Coordinate Geometry Basics**

Q105 Challenge

A robotic welding arm is programmed at point A $(-3,\quad 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point R symmetric with respect to the origin.

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(3,\quad -4)$.

Final answer

$(3,\quad -4)$

Worked Solutions **Coordinate Geometry Basics**

Q106 **Challenge**

A robotic welding arm is programmed at point A $(-3,\ 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-3,\ -4)$.

Correct option: A

Final answer

$(-3,\ -4)$

Worked Solutions **Coordinate Geometry Basics**

Q107 **Challenge**

A robotic welding arm is programmed at point A $(-3,\ 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point Q symmetric with respect to the y-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3,\ 4)$.

Correct option: B

Final answer

$(3,\ 4)$

Worked Solutions **Coordinate Geometry Basics**

Q108 **Challenge**

A robotic welding arm is programmed at point A $(-3,\ \ 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3,\ \ -4)$.

Correct option: C

Final answer

$(3,\ \ -4)$

Worked Solutions **Coordinate Geometry Basics**

Q109 Written

Distance between A $(2,\ \ 6)$ and B $(5,\ \ 10)$

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives 5.

Final answer

5

Q110 Written

Distance between A $(4,\ \ 3)$ and B $(1,\ \ 2)$

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $\sqrt{10}$.

Final answer

$\sqrt{10}$

Worked Solutions **Coordinate Geometry Basics**

Q111 **Written**

Distance between A($\sqrt{3}$,-1) and B $(0,\ \ 1)$

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $\sqrt{7}$.

Final answer

$\sqrt{7}$

Q112 **Written**

Distance between A $(1,\ \ 2)$ and B $(11,\ \ 7)$

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $5\sqrt{5}$.

Final answer

$5\sqrt{5}$

Worked Solutions **Coordinate Geometry Basics**

Q113 Written

Distance between A $(3, \ -1)$ and B $(-1, \ 3)$

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $4\sqrt{2}$.

Final answer

$4\sqrt{2}$

Q114 MCQ

Find distance between A $(2, \ 6)$ and B $(5, \ 10)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: C

Final answer

5

Worked Solutions **Coordinate Geometry Basics**

Q115 *MCQ*

Find distance between A $(4,\ 3)$ **and B** $(1,\ 2)$**:**

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $\sqrt{10}$.

Correct option: A

Final answer

$\sqrt{10}$

Q116 *MCQ*

Find distance between A($\sqrt{3}$**,-1) and B** $(0,\ 1)$**:**

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $\sqrt{7}$.

Correct option: C

Final answer

$\sqrt{7}$

Worked Solutions **Coordinate Geometry Basics**

Q117 MCQ

Find distance between A $(1,\ \ 2)$ and B $(11,\ \ 7)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $5\sqrt{5}$.

Correct option: B

Final answer

$5\sqrt{5}$

Q118 MCQ

Find distance between A $(3,\ \ -1)$ and B $(-1,\ \ 3)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $4\sqrt{2}$.

Correct option: A

Final answer

$4\sqrt{2}$

Worked Solutions **Coordinate Geometry Basics**

Q119 Written

A $(3, \quad -2)$, B $(x, \quad 1)$, distance 5: find x

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -1,\ 7$.

Final answer

$x = -1,\ 7$

Q120 Written

A $(-1, \quad 3)$, B $(x, \quad 1)$, distance $\sqrt{5}$: find x

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -2,\ 0$.

Final answer

$x = -2,\ 0$

Worked Solutions **Coordinate Geometry Basics**

Q121 Written

A $(3,\ -1)$, B $(x,\ 2)$, distance 5: find x

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -1,\ 7$.

Final answer

$x = -1,\ 7$

Q122 Written

A $(1,\ -3)$, B $(-2,\ y)$, distance $\sqrt{13}$: find y

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $y = -5,\ -1$.

Final answer

$y = -5,\ -1$

Worked Solutions **Coordinate Geometry Basics**

Q123 Written

A $(-4, \ -2)$, B $(x, \ 3)$, distance 5sqrt(5): find x

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -14, \ 6$.

Final answer

$x = -14, \ 6$

Q124 MCQ

A $(3, \ -2)$, B $(x, \ 1)$, distance 5: find x:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -1, \ 7$.

Correct option: B

Final answer

$x = -1, \ 7$

Q125 MCQ

A $(-1,\ 3)$, B $(x,\ 1)$, distance $\sqrt{5}$: find x:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -2,\ 0$.

Correct option: C

Final answer

$x = -2,\ 0$

Q126 MCQ

A $(3,\ -1)$, B $(x,\ 2)$, distance 5: find x:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -1,\ 7$.

Correct option: B

Final answer

$x = -1,\ 7$

Worked Solutions **Coordinate Geometry Basics**

Q127 MCQ

A $(1, \quad -3)$, B $(-2, \quad y)$, distance $\sqrt{13}$: find y:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $y = -5, \ -1$.

Correct option: A

Final answer

$y = -5, \ -1$

Q128 MCQ

A $(-4, \quad -2)$, B $(x, \quad 3)$, distance 5sqrt(5): find x:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -14, \ 6$.

Correct option: D

Final answer

$x = -14, \ 6$

Worked Solutions **Coordinate Geometry Basics**

Q129 **Written**

Distance from origin to A $(2, \ -4)$

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $2\sqrt{5}$.

Final answer

$2\sqrt{5}$

Q130 **Written**

Distance from origin to A $(-3, \ 4)$

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives 5.

Final answer

5

Worked Solutions **Coordinate Geometry Basics**

Q131 MCQ

Find distance from origin to A $(2,\ \ -4)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $2\sqrt{5}$.

Correct option: C

Final answer

$2\sqrt{5}$

Q132 MCQ

Find distance from origin to A $(-3,\ \ 4)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: C

Final answer

5

Worked Solutions **Coordinate Geometry Basics**

Q133 Challenge

Distance between (x,7) and (3x-1,-5) is 13: find x

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $x = -2,\ 3$.

Final answer

$x = -2,\ 3$

Worked Solutions **Coordinate Geometry Basics**

Q134 Challenge

Distance between (x,7) and (3x-1,-5) is 13: find x:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -2,\ 3$.

Correct option: C

Final answer

$x = -2,\ 3$

Worked Solutions **Coordinate Geometry Basics**

Q135 Written

Point on the x-axis equidistant from A $(-2,\ 1)$ and B $(3,\ 4)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(2,\ 0)$.

Final answer

$(2,\ 0)$

Q136 Written

Point on the y-axis equidistant from A $(4,\ 2)$ and B $(1,\ -1)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(0,\ 3)$.

Final answer

$(0,\ 3)$

Worked Solutions **Coordinate Geometry Basics**

Q137 Written

Point on the x-axis equidistant from A $(-1, \ 2)$ and B $(4, \ 3)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(2, \ 0)$.

Final answer

$(2, \ 0)$

Q138 Written

Point on the y-axis equidistant from A $(-5, \ 2)$ and B $(3, \ 5)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(0, \ 5/6)$.

Final answer

$(0, \ 5/6)$

Worked Solutions **Coordinate Geometry Basics**

Q139 Written

Point on the x-axis equidistant from A $(-1, \; 2)$ and B $(3, \; 8)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(17/2, \; 0)$.

Final answer

$(17/2, \; 0)$

Q140 MCQ

Find point on the x-axis equidistant from A $(-2, \; 1)$ and B $(3, \; 4)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, \; 0)$.

Correct option: C

Final answer

$(2, \; 0)$

Worked Solutions **Coordinate Geometry Basics**

Q141 MCQ

Find point on the y-axis equidistant from A $(4,\ 2)$ and B $(1,\ -1)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0,\ 3)$.

Correct option: A

Final answer

$(0,\ 3)$

Q142 MCQ

Find point on the x-axis equidistant from A $(-1,\ 2)$ and B $(4,\ 3)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2,\ 0)$.

Correct option: C

Final answer

$(2,\ 0)$

Worked Solutions **Coordinate Geometry Basics**

Q143 MCQ

Find point on the y-axis equidistant from A $(-5,\ 2)$ and B $(3,\ 5)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0,\ 5/6)$.

Correct option: C

Final answer

$(0,\ 5/6)$

Q144 MCQ

Find point on the x-axis equidistant from A $(-1,\ 2)$ and B $(3,\ 8)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(17/2,\ 0)$.

Correct option: B

Final answer

$(17/2,\ 0)$

Worked Solutions **Coordinate Geometry Basics**

Q145 Written

Point on $y = 2x - 1$ equidistant from A $(1,\ \ 4)$ and B $(3,\ \ 1)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(13/8,\ \ 9/4)$.

Final answer

$(13/8,\ \ 9/4)$

Q146 Written

Point on $y = 3x + 1$ equidistant from A $(4,\ \ -5)$ and B $(2,\ \ -3)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(-4,\ \ -11)$.

Final answer

$(-4,\ \ -11)$

Worked Solutions **Coordinate Geometry Basics**

Q147 Written

Point on $y = 2x - 3$ equidistant from A $(1,\ \ 4)$ and B $(3,\ \ -2)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(2,\ \ 1)$.

Final answer

$(2,\ \ 1)$

Q148 Written

Point on $y = x - 1$ equidistant from A $(-1,\ \ 5)$ and B $(7,\ \ -1)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(3,\ \ 2)$.

Final answer

$(3,\ \ 2)$

Worked Solutions **Coordinate Geometry Basics**

Q149 Written

Point on $y = -x + 1$ equidistant from A $(1,\ 2)$ and B $(-3,\ 6)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(-2,\ 3)$.

Final answer

$(-2,\ 3)$

Q150 MCQ

Find point on $y = 2x - 1$ equidistant from A $(1,\ 4)$ and B $(3,\ 1)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(13/8,\ 9/4)$.

Correct option: B

Final answer

$(13/8,\ 9/4)$

Worked Solutions **Coordinate Geometry Basics**

Q151 MCQ

Find point on $y = 3x + 1$ equidistant from A $(4,\ -5)$ and B $(2,\ -3)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-4,\ -11)$.

Correct option: C

Final answer

$(-4,\ -11)$

Q152 MCQ

Find point on $y = 2x - 3$ equidistant from A $(1,\ 4)$ and B $(3,\ -2)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2,\ 1)$.

Correct option: A

Final answer

$(2,\ 1)$

Worked Solutions **Coordinate Geometry Basics**

Q153 MCQ

Find point on $y = x - 1$ equidistant from A $(-1,\ \ 5)$ and B $(7,\ \ -1)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3,\ \ 2)$.

Correct option: C

Final answer

$(3,\ \ 2)$

Q154 MCQ

Find point on $y = -x + 1$ equidistant from A $(1,\ \ 2)$ and B $(-3,\ \ 6)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-2,\ \ 3)$.

Correct option: B

Final answer

$(-2,\ \ 3)$

Worked Solutions **Coordinate Geometry Basics**

Q155 Challenge

A $(2, \ 1)$, B $(4, \ 5)$, C $(3, \ y)$ with CA=CB and D $(x, \ 0)$ with DA=DB: find CD

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $3\sqrt{5}$.

Final answer

$3\sqrt{5}$

Worked Solutions **Coordinate Geometry Basics**

Q156 Challenge

A $(2,\ \ 1)$, B $(4,\ \ 5)$, C $(3,\ \ y)$ with CA=CB and D $(x,\ \ 0)$ with DA=DB: find CD:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $3\sqrt{5}$.

Correct option: A

Final answer

$3\sqrt{5}$

Worked Solutions **Coordinate Geometry Basics**

Q157 Written

Given A $(-2,\ -3)$, B $(3,\ 7)$, and C $(5,\ 2)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(4/3,\ 11/3)$.

Final answer

$(4/3,\ 11/3)$

Worked Solutions **Coordinate Geometry Basics**

Q158 Written

Given A $(-3,\ 4)$, B $(1,\ -4)$, and C $(-4,\ 3)$, find the coordinates of point P dividing segment AC internally in the ratio 3:1.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-15/4,\ 13/4)$.

Final answer

$(-15/4,\ 13/4)$

Worked Solutions **Coordinate Geometry Basics**

Q159 Written

Given A $(2,\ 3)$, B $(2,\ -3)$, and C $(-2,\ 3)$, find the coordinates of point P dividing segment AB internally in the ratio 4:1.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2,\ -9/5)$.

Final answer

$(2,\ -9/5)$

Worked Solutions **Coordinate Geometry Basics**

Q160 Written

Given A $(-2,\ 1)$, B $(6,\ -3)$, and C $(1,\ 7)$, find the coordinates of point P dividing segment BC internally in the ratio 3:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(3,\ 3)$.

Final answer

$(3,\ 3)$

Worked Solutions **Coordinate Geometry Basics**

Q161 Written

Given A $(-4, \ 5)$, B $(1, \ -5)$, and C $(3, \ 0)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-2/3, \ -5/3)$.

Final answer

$(-2/3, \ -5/3)$

Worked Solutions **Coordinate Geometry Basics**

Q162 MCQ

Given A $(-2,\ -3)$, B $(3,\ 7)$, and C $(5,\ 2)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4/3,\ 11/3)$.

Correct option: C

Final answer

$(4/3,\ 11/3)$

Worked Solutions **Coordinate Geometry Basics**

Q163 MCQ

Given A $(-3, \ 4)$, B $(1, \ -4)$, and C $(-4, \ 3)$, find the coordinates of point P dividing segment AC internally in the ratio 3:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-15/4, \ 13/4)$.

Correct option: C

Final answer

$(-15/4, \ 13/4)$

Worked Solutions **Coordinate Geometry Basics**

Q164 MCQ

Given A $(2,\ 3)$, B $(2,\ -3)$, and C $(-2,\ 3)$, find the coordinates of point P dividing segment AB internally in the ratio 4:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2,\ -9/5)$.

Correct option: C

Final answer

$(2,\ -9/5)$

Worked Solutions **Coordinate Geometry Basics**

Q165 MCQ

Given A $(-2,\ 1)$, B $(6,\ -3)$, and C $(1,\ 7)$, find the coordinates of point P dividing segment BC internally in the ratio 3:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3,\ 3)$.

Correct option: C

Final answer

$(3,\ 3)$

Worked Solutions **Coordinate Geometry Basics**

Q166 MCQ

Given A $(-4,\ 5)$, B $(1,\ -5)$, and C $(3,\ 0)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-2/3,\ -5/3)$.

Correct option: A

Final answer

$(-2/3,\ -5/3)$

Worked Solutions **Coordinate Geometry Basics**

Q167 Written

Given A $(-2,\ -3)$, B $(3,\ 7)$, and C $(5,\ 2)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2,\ 19/2)$.

Final answer

$(2,\ 19/2)$

Worked Solutions **Coordinate Geometry Basics**

Q168 Written

Given A $(-3,\ 4)$, B $(1,\ -4)$, and C $(-4,\ 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(11,\ -18)$.

Final answer

$(11,\ -18)$

Worked Solutions **Coordinate Geometry Basics**

Q169 Written

Given A $(2,\ 3)$, B $(2,\ -3)$, and C $(-2,\ 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(10,\ -15)$.

Final answer

$(10,\ -15)$

Worked Solutions **Coordinate Geometry Basics**

Q170 Written

Given A $(-2,\ 1)$, B $(6,\ -3)$, and C $(1,\ 7)$, find the coordinates of point Q dividing segment CA externally in the ratio 3:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-8,\ -11)$.

Final answer

$(-8,\ -11)$

Worked Solutions **Coordinate Geometry Basics**

Q171 Written

Given A $(-4,\ 5)$, B $(1,\ -5)$, and C $(3,\ 0)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(0,\ -15/2)$.

Final answer

$(0,\ -15/2)$

Worked Solutions **Coordinate Geometry Basics**

Q172 MCQ

Given A $(-2, \ -3)$, B $(3, \ 7)$, and C $(5, \ 2)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, \ 19/2)$.

Correct option: B

Final answer

$(2, \ 19/2)$

Worked Solutions **Coordinate Geometry Basics**

Q173 MCQ

Given A $(-3,\ 4)$, B $(1,\ -4)$, and C $(-4,\ 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(11,\ -18)$.

Correct option: C

Final answer

$(11,\ -18)$

Worked Solutions **Coordinate Geometry Basics**

Q174 MCQ

Given A $(2,\ 3)$, B $(2,\ -3)$, and C $(-2,\ 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(10,\ -15)$.

Correct option: B

Final answer

$(10,\ -15)$

Worked Solutions **Coordinate Geometry Basics**

Q175 MCQ

Given A $(-2,\ 1)$, B $(6,\ -3)$, and C $(1,\ 7)$, find the coordinates of point Q dividing segment CA externally in the ratio 3:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-8,\ -11)$.

Correct option: B

Final answer

$(-8,\ -11)$

Worked Solutions **Coordinate Geometry Basics**

Q176 MCQ

Given A $(-4,\ 5)$, B $(1,\ -5)$, and C $(3,\ 0)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0,\ -15/2)$.

Correct option: C

Final answer

$(0,\ -15/2)$

Worked Solutions **Coordinate Geometry Basics**

Q177 Written

Given A $(-2, \ -3)$, B $(3, \ 7)$, and C $(5, \ 2)$, find the coordinates of midpoint M of segment CA.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(3/2, \ -1/2)$.

Final answer

$(3/2, \ -1/2)$

Worked Solutions **Coordinate Geometry Basics**

Q178 Written

Given A $(-2,\ -3)$, B $(3,\ 7)$, and C $(5,\ 2)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2,\ 2)$.

Final answer

$(2,\ 2)$

Worked Solutions **Coordinate Geometry Basics**

Q179 Written

Given A $(-3,\ 4)$, B $(1,\ -4)$, and C $(-4,\ 3)$, find the coordinates of midpoint M of segment AB.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-1,\ 0)$.

Final answer

$(-1,\ 0)$

Worked Solutions **Coordinate Geometry Basics**

Q180 Written

Given A $(-3,\ 4)$, B $(1,\ -4)$, and C $(-4,\ 3)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-2,\ 1)$.

Final answer

$(-2,\ 1)$

Worked Solutions **Coordinate Geometry Basics**

Q181 Written

Given A $(2, \quad 3)$, B $(2, \quad -3)$, and C $(-2, \quad 3)$, find the coordinates of midpoint M of segment CA.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(0, \quad 3)$.

Final answer

$(0, \quad 3)$

Worked Solutions **Coordinate Geometry Basics**

Q182 Written

Given A $(2,\ 3)$, B $(2,\ -3)$, and C $(-2,\ 3)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2/3,\ 1)$.

Final answer

$(2/3,\ 1)$

Worked Solutions **Coordinate Geometry Basics**

Q183 Written

Given A $(-2,\ 1)$, B $(6,\ -3)$, and C $(1,\ 7)$, find the coordinates of midpoint M of segment AB.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2,\ -1)$.

Final answer

$(2,\ -1)$

Worked Solutions **Coordinate Geometry Basics**

Q184 **Written**

Given A $(-2,\ 1)$, B $(6,\ -3)$, and C $(1,\ 7)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(5/3,\ 5/3)$.

Final answer

$(5/3,\ 5/3)$

Worked Solutions **Coordinate Geometry Basics**

Q185 Written

Given A $(-4, \quad 5)$, B $(1, \quad -5)$, and C $(3, \quad 0)$, find the coordinates of midpoint M of segment BC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2, \quad -5/2)$.

Final answer

$(2, \quad -5/2)$

Worked Solutions **Coordinate Geometry Basics**

Q186 Written

Given A $(-4,\ 5)$, B $(1,\ -5)$, and C $(3,\ 0)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(0,\ 0)$.

Final answer

$(0,\ 0)$

Worked Solutions **Coordinate Geometry Basics**

Q187 MCQ

Given A $(-2, \ -3)$, B $(3, \ 7)$, and C $(5, \ 2)$, find the coordinates of midpoint M of segment CA:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3/2, \ -1/2)$.

Correct option: A

Final answer

$(3/2, \ -1/2)$

Worked Solutions **Coordinate Geometry Basics**

Q188 MCQ

Given A $(-2,\ -3)$, B $(3,\ 7)$, and C $(5,\ 2)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2,\ 2)$.

Correct option: C

Final answer

$(2,\ 2)$

Worked Solutions **Coordinate Geometry Basics**

Q189 MCQ

Given A $(-3,\ 4)$, B $(1,\ -4)$, and C $(-4,\ 3)$, find the coordinates of midpoint M of segment AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-1,\ 0)$.

Correct option: B

Final answer

$(-1,\ 0)$

Worked Solutions **Coordinate Geometry Basics**

Q190 MCQ

Given A $(-3,\ 4)$, B $(1,\ -4)$, and C $(-4,\ 3)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-2,\ 1)$.

Correct option: A

Final answer

$(-2,\ 1)$

Worked Solutions **Coordinate Geometry Basics**

Q191 MCQ

Given A $(2,\ 3)$, B $(2,\ -3)$, and C $(-2,\ 3)$, find the coordinates of midpoint M of segment CA:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0,\ 3)$.

Correct option: A

Final answer

$(0,\ 3)$

Worked Solutions **Coordinate Geometry Basics**

Q192 MCQ

Given A $(2, \ 3)$, B $(2, \ -3)$, and C $(-2, \ 3)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2/3, \ 1)$.

Correct option: D

Final answer

$(2/3, \ 1)$

Worked Solutions **Coordinate Geometry Basics**

Q193 MCQ

Given A $(-2,\ 1)$, B $(6,\ -3)$, and C $(1,\ 7)$, find the coordinates of midpoint M of segment AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2,\ -1)$.

Correct option: B

Final answer

$(2,\ -1)$

Worked Solutions **Coordinate Geometry Basics**

Q194 MCQ

Given A $(-2,\ 1)$, B $(6,\ -3)$, and C $(1,\ 7)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(5/3,\ 5/3)$.

Correct option: A

Final answer

$(5/3,\ 5/3)$

Worked Solutions **Coordinate Geometry Basics**

Q195 MCQ

Given A $(-4,\ 5)$, B $(1,\ -5)$, and C $(3,\ 0)$, find the coordinates of midpoint M of segment BC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2,\ -5/2)$.

Correct option: B

Final answer

$(2,\ -5/2)$

Worked Solutions Coordinate Geometry Basics

Q196 MCQ

Given A $(-4,\ 5)$, B $(1,\ -5)$, and C $(3,\ 0)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0,\ 0)$.

Correct option: C

Final answer

$(0,\ 0)$

Worked Solutions **Coordinate Geometry Basics**

Q197 Challenge

A $(-2, \quad 3)$, B $(6, \quad 1)$, C $(0, \quad y)$, CA=CB: find the centroid G

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(4/3, \quad -2/3)$.

Final answer

$(4/3, \quad -2/3)$

Worked Solutions **Coordinate Geometry Basics**

Q198 Challenge

A $(-2,\quad 3)$**, B** $(6,\quad 1)$**, C** $(0,\quad y)$**, CA=CB: find the centroid G:**

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4/3,\quad -2/3)$.

Correct option: C

Final answer

$(4/3,\quad -2/3)$

Worked Solutions **Coordinate Geometry Basics**

Q199 Written

Point with A $(-3,\ \ 2)$ symmetric about centre (2, 6)

Solution

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(7,\ \ 10)$.

Final answer

$(7,\ \ 10)$

Q200 Written

Point with A $(1,\ \ -2)$ symmetric about centre (2, 3)

Solution

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(3,\ \ 8)$.

Final answer

$(3,\ \ 8)$

Worked Solutions **Coordinate Geometry Basics**

Q201 Written

Point with A $(3,\ \ 2)$ symmetric about centre (-1, 5)

Solution

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-5,\ \ 8)$.

Final answer

$(-5,\ \ 8)$

Q202 Written

Point with A $(-3,\ \ 1)$ symmetric about centre (2, 0)

Solution

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(7,\ \ -1)$.

Final answer

$(7,\ \ -1)$

Q203 Written

Point with A $(7, \quad 4)$ symmetric about centre (-3, 5)

Solution

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-13, \quad 6)$.

Final answer

$(-13, \quad 6)$

Q204 Written

Point with A $(4, \quad -8)$ symmetric about centre (-3, 2)

Solution

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-10, \quad 12)$.

Final answer

$(-10, \quad 12)$

Worked Solutions **Coordinate Geometry Basics**

Q205 Written

Point with A $(0,\ -4)$ symmetric about centre (-4, 1)

Solution

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-8,\ 6)$.

Final answer

$(-8,\ 6)$

Q206 Written

Point with A $(-4,\ 2)$ symmetric about centre (-1, -3)

Solution

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(2,\ -8)$.

Final answer

$(2,\ -8)$

Worked Solutions · **Coordinate Geometry Basics**

Q207 MCQ

Find point with A $(-3,\ 2)$ symmetric about centre (2, 6):

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(7,\ 10)$.

Correct option: C

Final answer

$(7,\ 10)$

Q208 MCQ

Find point with A $(1,\ -2)$ symmetric about centre (2, 3):

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3,\ 8)$.

Correct option: B

Final answer

$(3,\ 8)$

Worked Solutions **Coordinate Geometry Basics**

Q209 MCQ

Find point with A $(3,\ 2)$ symmetric about centre (-1, 5):

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-5,\ 8)$.

Correct option: A

Final answer

$(-5,\ 8)$

Q210 MCQ

Find point with A $(-3,\ 1)$ symmetric about centre (2, 0):

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(7,\ -1)$.

Correct option: C

Final answer

$(7,\ -1)$

Worked Solutions **Coordinate Geometry Basics**

Q211 MCQ

Find point with A $(7,\ \ 4)$ symmetric about centre (-3, 5):

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-13,\ \ 6)$.

Correct option: C

Final answer

$(-13,\ \ 6)$

Q212 MCQ

Find point with A $(4,\ \ -8)$ symmetric about centre (-3, 2):

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-10,\ \ 12)$.

Correct option: C

Final answer

$(-10,\ \ 12)$

Worked Solutions **Coordinate Geometry Basics**

Q213 MCQ

Find point with A $(0,\ -4)$ symmetric about centre (-4, 1):

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-8,\ 6)$.

Correct option: B

Final answer

$(-8,\ 6)$

Q214 MCQ

Find point with A $(-4,\ 2)$ symmetric about centre (-1, -3):

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2,\ -8)$.

Correct option: A

Final answer

$(2,\ -8)$

Worked Solutions **Coordinate Geometry Basics**

Q215 Challenge

B $(m,\ \ 5)$, C $(4,\ \ n)$ symmetric about A $(1,\ \ -1)$: find m+n

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is -9.

Final answer

-9

Worked Solutions | **Coordinate Geometry Basics**

Q216 Challenge

B $(m, \ 5)$, C $(4, \ n)$ symmetric about A $(1, \ -1)$: find m+n:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -9.

Correct option: C

Final answer

-9

Worked Solutions **Coordinate Geometry Basics**

Quiz MCQ 1 **Concept Quiz · MCQ**

Point A $(-3,\ \ 4)$ lies in which quadrant?:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is II.

Correct option: A

Final answer

II

Quiz MCQ 2 **Concept Quiz · MCQ**

Find the distance between A $(1,\ \ 2)$ and B $(4,\ \ 6)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: B

Final answer

5

Worked Solutions **Coordinate Geometry Basics**

Quiz MCQ 3 **Concept Quiz · MCQ**

Point P divides the segment from A $(-2,\ 3)$ to B $(6,\ 1)$ internally in ratio 1:3. Find P:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0,\ 5/2)$.

Correct option: C

Final answer

$(0,\ 5/2)$

Quiz MCQ 4 **Concept Quiz · MCQ**

Point B is symmetric to A $(2,\ -1)$ about P $(-3,\ 4)$. Find B:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-8,\ 9)$.

Correct option: C

Final answer

$(-8,\ 9)$

Worked Solutions **Coordinate Geometry Basics**

Quiz WRITTEN 5 Concept Quiz · Written

Find the point on $y = 2x + 1$ equidistant from A $(0,\ \ 2)$ and B $(4,\ \ 0)$.

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(1,\ \ 3)$.

Final answer

$(1,\ \ 3)$

Quiz WRITTEN 6 Concept Quiz · Written

For points A (-5) and B (7), find the point dividing AB externally in ratio 2:5.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives -13.
3. Final answer: -13.

Final answer

-13

Worked Solutions **Coordinate Geometry Basics**

Quiz WRITTEN 7 Concept Quiz · Written

If the distance between A $(3, \ -1)$ and B $(x, \ 2)$ is 5, find x.

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -1, \ 7$.

Final answer

$x = -1, \ 7$

Quiz WRITTEN 8 Concept Quiz · Written

Find the centroid of A $(-2, \ 1)$, B $(6, \ -3)$, C $(1, \ 7)$.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(5/3, \ 5/3)$.

Final answer

$(5/3, \ 5/3)$

Answer key

Use the question number shown on each page.

Q001 12
Q002 12
Q003 8
Q004 6
Q005 C
Q006 C
Q007 C
Q008 C
Q009 4
Q010 3
Q011 4
Q012 8
Q013 B
Q014 C
Q015 B
Q016 B
Q017 -28
Q018 -13
Q019 18
Q020 1
Q021 A
Q022 B
Q023 A
Q024 B
Q025 2
Q026 1
Q027 2
Q028 7
Q029 C
Q030 A
Q031 D
Q032 A
Q033 5
Q034 -7
Q035 -23
Q036 $-11/2$
Q037 A
Q038 C
Q039 B
Q040 C
Q041 I
Q042 IV
Q043 II
Q044 III
Q045 II
Q046 I
Q047 IV
Q048 III
Q049 II
Q050 IV
Q051 I
Q052 III
Q053 II
Q054 IV
Q055 III
Q056 I
Q057 C
Q058 B
Q059 A
Q060 C
Q061 A
Q062 C
Q063 B
Q064 A
Q065 B
Q066 A
Q067 A
Q068 C
Q069 B
Q070 C
Q071 C
Q072 D
Q073 $(-3,\ -2)$
Q074 $(-1,\ -2)$
Q075 $(-4,\ 2)$
Q076 $(1,\ 2)$
Q077 $(-3,\ -4)$
Q078 C
Q079 D
Q080 C
Q081 D
Q082 C
Q083 $(3,\ 2)$
Q084 $(1,\ 2)$
Q085 $(4,\ -2)$
Q086 $(-1,\ -2)$
Q087 $(3,\ 4)$
Q088 C
Q089 C
Q090 A
Q091 A
Q092 A
Q093 $(3,\ -2)$
Q094 $(1,\ -2)$
Q095 $(4,\ 2)$
Q096 $(-1,\ 2)$
Q097 $(3,\ -4)$
Q098 B
Q099 B
Q100 B
Q101 A
Q102 D
Q103 $(-3,\ -4)$
Q104 $(3,\ 4)$

Q105 $(3,\ -4)$
Q106 A
Q107 B
Q108 C
Q109 5
Q110 $\sqrt{10}$
Q111 $\sqrt{7}$
Q112 $5\sqrt{5}$
Q113 $4\sqrt{2}$
Q114 C
Q115 A
Q116 C
Q117 B
Q118 A
Q119 $x = -1,\ 7$
Q120 $x = -2,\ 0$
Q121 $x = -1,\ 7$
Q122 $y = -5,\ -1$
Q123 $x = -14,\ 6$
Q124 B
Q125 C
Q126 B
Q127 A
Q128 D
Q129 $2\sqrt{5}$
Q130 5
Q131 C
Q132 C
Q133 $x = -2,\ 3$
Q134 C
Q135 $(2,\ 0)$
Q136 $(0,\ 3)$
Q137 $(2,\ 0)$
Q138 $(0,\ 5/6)$
Q139 $(17/2,\ 0)$
Q140 C
Q141 A
Q142 C
Q143 C
Q144 B
Q145 $(13/8,\ 9/4)$
Q146 $(-4,\ -11)$
Q147 $(2,\ 1)$
Q148 $(3,\ 2)$
Q149 $(-2,\ 3)$
Q150 B
Q151 C
Q152 A
Q153 C
Q154 B
Q155 $3\sqrt{5}$
Q156 A
Q157 $(4/3,\ 11/3)$
Q158 $(-15/4,\ 13/4)$
Q159 $(2,\ -9/5)$
Q160 $(3,\ 3)$
Q161 $(-2/3,\ -5/3)$
Q162 C
Q163 C
Q164 C
Q165 C
Q166 A
Q167 $(2,\ 19/2)$
Q168 $(11,\ -18)$
Q169 $(10,\ -15)$
Q170 $(-8,\ -11)$
Q171 $(0,\ -15/2)$
Q172 B
Q173 C
Q174 B
Q175 B
Q176 C
Q177 $(3/2,\ -1/2)$
Q178 $(2,\ 2)$
Q179 $(-1,\ 0)$
Q180 $(-2,\ 1)$
Q181 $(0,\ 3)$
Q182 $(2/3,\ 1)$
Q183 $(2,\ -1)$
Q184 $(5/3,\ 5/3)$
Q185 $(2,\ -5/2)$
Q186 $(0,\ 0)$
Q187 A
Q188 C
Q189 B
Q190 A
Q191 A
Q192 D
Q193 B
Q194 A
Q195 B
Q196 C
Q197 $(4/3,\ -2/3)$
Q198 C
Q199 $(7,\ 10)$
Q200 $(3,\ 8)$
Q201 $(-5,\ 8)$
Q202 $(7,\ -1)$
Q203 $(-13,\ 6)$
Q204 $(-10,\ 12)$
Q205 $(-8,\ 6)$
Q206 $(2,\ -8)$
Q207 C
Q208 B
Q209 A
Q210 C
Q211 C
Q212 C
Q213 B
Q214 A
Q215 -9
Q216 C
Quiz MCQ 1 A
Quiz MCQ 2 B
Quiz MCQ 3 C
Quiz MCQ 4 C
Quiz WRITTEN 5 $(1,\ 3)$
Quiz WRITTEN 6 -13
Quiz WRITTEN 7 $x = -1,\ 7$
Quiz WRITTEN 8 $(5/3,\ 5/3)$

Concept 2 | Triangles and Lines

English Student Edition • Focused formulas and guided practice

Formula opener

Line

$$y - y_1 = m(x - x_1)$$

Perpendicular

$$m_1 m_2 = -1$$

Area

$$A = \frac{1}{2}bh$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Triangles and Lines**

Q001 **Written**

Determine the shape of triangle ABC with A $(3,\ \ 2)$, B $(-1,\ \ 0)$, C $\left(1+\sqrt{3},\ \ 1-2\sqrt{3}\right)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2+b^2=c^2$, taking c as the longest side.
3. Final classification: *equilateral*.

Final answer

equilateral

Worked Solutions **Triangles and Lines**

Q002 **Written**

Determine the shape of triangle ABC with A $(2,\ \ 2)$, B $(2,\ \ 0)$, C $\left(2+\sqrt{3},\ \ 1\right)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2+b^2=c^2$, taking c as the longest side.
3. Final classification: *equilateral*.

Final answer

equilateral

Worked Solutions **Triangles and Lines**

Q003 *MCQ*

Determine the shape of triangle ABC with A $(3,\ \ 2)$, B $(-1,\ \ 0)$, C $\left(1+\sqrt{3},\ \ 1-2\sqrt{3}\right)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *equilateral*.

Correct option: D

Final answer

equilateral

Worked Solutions   **Triangles and Lines**

Q004 MCQ

Determine the shape of triangle ABC with A $(2,\ \ 2)$, B $(2,\ \ 0)$, C $\left(2+\sqrt{3},\ \ 1\right)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $equilateral$.

Correct option: C

Final answer

$equilateral$

Q005 Written

Determine the shape of triangle ABC with A $(-1,\ \ 2)$, B $(1,\ \ 1)$, C $(0,\ \ 4)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: $isosceles\ right - angled;\ right\ angle\ at\ A$.

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ A$

Worked Solutions **Triangles and Lines**

Q006 Written

Determine the shape of triangle ABC with A $(0,\ \ 2)$, B $(-1,\ \ -2)$, C $(4,\ \ 1)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: $isosceles\ right-angled;\ right\ angle\ at\ A$.

Final answer

$isosceles\ right-angled;\ right\ angle\ at\ A$

Q007 MCQ

Determine the shape of triangle ABC with A $(-1,\ \ 2)$, B $(1,\ \ 1)$, C $(0,\ \ 4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $isosceles\ right-angled;\ right\ angle\ at\ A$.

Correct option: B

Final answer

$isosceles\ right-angled;\ right\ angle\ at\ A$

Worked Solutions **Triangles and Lines**

Q008 MCQ

Determine the shape of triangle ABC with A $(0,\ \ 2)$, B $(-1,\ \ -2)$, C $(4,\ \ 1)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $isosceles\ right - angled;\ right\ angle\ at\ A$.

Correct option: D

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ A$

Q009 Written

Determine the shape of triangle ABC with A $(-3,\ \ 4)$, B $(1,\ \ 7)$, C $(3,\ \ -4)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: $scalene\ right - angled;\ right\ angle\ at\ A$.

Final answer

$scalene\ right - angled;\ right\ angle\ at\ A$

Worked Solutions Triangles and Lines

Q010 MCQ

Determine the shape of triangle ABC with A $(-3,\ \ 4)$, B $(1,\ \ 7)$, C $(3,\ \ -4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $scalene\ right - angled;\ right\ angle\ at\ A$.

Correct option: D

Final answer

$scalene\ right - angled;\ right\ angle\ at\ A$

Q011 Written

Determine the shape of triangle ABC with A $(4,\ \ 9)$, B $(0,\ \ 6)$, C $(3,\ \ 2)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: $isosceles\ right - angled;\ right\ angle\ at\ B$.

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ B$

Worked Solutions **Triangles and Lines**

Q012 **Written**

Determine the shape of triangle ABC with A $(1,\ \ 2)$, B $(5,\ \ 8)$, C $(11,\ \ 4)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: $isosceles\ right - angled;\ right\ angle\ at\ B$.

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ B$

Q013 **Written**

Determine the shape of triangle ABC with A $(-1,\ \ 1)$, B $(4,\ \ 2)$, C $(1,\ \ 4)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: $isosceles\ right - angled;\ right\ angle\ at\ C$.

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ C$

Worked Solutions **Triangles and Lines**

Q014 MCQ

Determine the shape of triangle ABC with A $(4,\ 9)$, B $(0,\ 6)$, C $(3,\ 2)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $isosceles\ right-angled;\ right\ angle\ at\ B$.

Correct option: B

Final answer

$isosceles\ right-angled;\ right\ angle\ at\ B$

Q015 MCQ

Determine the shape of triangle ABC with A $(1,\ 2)$, B $(5,\ 8)$, C $(11,\ 4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $isosceles\ right-angled;\ right\ angle\ at\ B$.

Correct option: C

Final answer

$isosceles\ right-angled;\ right\ angle\ at\ B$

Worked Solutions **Triangles and Lines**

Q016 MCQ

Determine the shape of triangle ABC with A $(-1,\ 1)$, B $(4,\ 2)$, C $(1,\ 4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $isosceles\ right - angled;\ right\ angle\ at\ C$.

Correct option: B

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ C$

Worked Solutions **Triangles and Lines**

Q017 **Challenge**

If A $(1,\ \ 2)$, B $(2k,\ \ k)$, C $(4,\ \ 5)$ and ABC is isosceles and right-angled at B, find k.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $k = 2$.

Final answer

$k = 2$

Worked Solutions **Triangles and Lines**

Q018 Challenge

If A $(1,\ 2)$, B $(2k,\ k)$, C $(4,\ 5)$ and ABC is isosceles and right-angled at B, find k:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $k = 2$.

Correct option: C

Final answer

$k = 2$

Worked Solutions **Triangles and Lines**

Q019 Written

Find the equation of the line through (3,-4) parallel to the x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -4$.

Final answer

$y = -4$

Q020 Written

Find the equation of the line through (-3,8) parallel to the y-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -3$.

Final answer

$x = -3$

Worked Solutions **Triangles and Lines**

Q021 Written

Find the equation of the line through (2,6) parallel to the x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 6$.

Final answer

$y = 6$

Q022 Written

Find the equation of the line through (3,5) perpendicular to the x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 3$.

Final answer

$x = 3$

Worked Solutions **Triangles and Lines**

Q023 **Written**

Find the equation of the line through (-4,5) parallel to x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 5$.

Final answer

$y = 5$

Q024 **Written**

Find the equation of the line through (2,1) parallel to x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 1$.

Final answer

$y = 1$

Worked Solutions **Triangles and Lines**

Q025 Written

Find the equation of the line through (-4,-7) parallel to x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -7$.

Final answer

$y = -7$

Q026 Written

Find the equation of the line through (1,-2) perpendicular to x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 1$.

Final answer

$x = 1$

Worked Solutions **Triangles and Lines**

Q027 **Written**

Find the equation of the line through (-3,-4) perpendicular to x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -3$.

Final answer

$x = -3$

Q028 **Written**

Find the equation of the line through (3,-5) perpendicular to x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 3$.

Final answer

$x = 3$

Worked Solutions **Triangles and Lines**

Q029 Written

Find the equation of the line through (2,-7) parallel to y-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 2$.

Final answer

$x = 2$

Q030 Written

Find the equation of the line through (4,3) parallel to y-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 4$.

Final answer

$x = 4$

Worked Solutions **Triangles and Lines**

Q031 Written

Find the equation of the line through (-6,-2) parallel to y-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -6$.

Final answer

$x = -6$

Q032 MCQ

Find the equation of the line through (3,-4) parallel to the x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -4$.

Correct option: B

Final answer

$y = -4$

Worked Solutions **Triangles and Lines**

Q033 MCQ

Find the equation of the line through (-3,8) parallel to the y-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -3$.

Correct option: C

Final answer

$x = -3$

Q034 MCQ

Find the equation of the line through (2,6) parallel to the x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 6$.

Correct option: D

Final answer

$y = 6$

Worked Solutions **Triangles and Lines**

Q035 MCQ

Find the equation of the line through (3,5) perpendicular to the x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 3$.

Correct option: D

Final answer

$x = 3$

Q036 MCQ

Find the equation of the line through (-4,5) parallel to x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 5$.

Correct option: B

Final answer

$y = 5$

Worked Solutions **Triangles and Lines**

Q037 MCQ

Find the equation of the line through (2,1) parallel to x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 1$.

Correct option: D

Final answer

$y = 1$

Q038 MCQ

Find the equation of the line through (-4,-7) parallel to x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -7$.

Correct option: B

Final answer

$y = -7$

Worked Solutions **Triangles and Lines**

Q039 *MCQ*

Find the equation of the line through (1,-2) perpendicular to x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 1$.

Correct option: C

Final answer

$x = 1$

Q040 *MCQ*

Find the equation of the line through (-3,-4) perpendicular to x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -3$.

Correct option: D

Final answer

$x = -3$

Worked Solutions **Triangles and Lines**

Q041 MCQ

Find the equation of the line through (3,-5) perpendicular to x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 3$.

Correct option: D

Final answer

$x = 3$

Q042 MCQ

Find the equation of the line through (2,-7) parallel to y-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 2$.

Correct option: B

Final answer

$x = 2$

Worked Solutions **Triangles and Lines**

Q043 MCQ

Find the equation of the line through (4,3) parallel to y-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 4$.

Correct option: D

Final answer

$x = 4$

Q044 MCQ

Find the equation of the line through (-6,-2) parallel to y-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -6$.

Correct option: A

Final answer

$x = -6$

Worked Solutions **Triangles and Lines**

Q045 **Written**

Find the equation of the line through (-3,1) and (-3,2).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -3$.

Final answer

$x = -3$

Q046 **Written**

Find the equation of the line through (-2,3) and (-5,3).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 3$.

Final answer

$y = 3$

Worked Solutions **Triangles and Lines**

Q047 Written

Find the equation of the line through (1,-1) and (4,0).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = x/3 - 4/3(\text{equivalently} x - 3y - 4 = 0)$.

Final answer

$y = x/3 - 4/3(\text{equivalently} x - 3y - 4 = 0)$

Q048 Written

Find the equation of the line through (-2,-2) and (2,-2).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -2$.

Final answer

$y = -2$

Worked Solutions **Triangles and Lines**

Q049 Written

Find the equation of the line through (-4,2) and (-4,8).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -4$.

Final answer

$x = -4$

Q050 Written

Find the equation of the line through (-2,-6) and (-4,1).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $7x + 2y + 26 = 0$.

Final answer

$7x + 2y + 26 = 0$

Worked Solutions **Triangles and Lines**

Q051 **Written**

Find the equation of the line through (4,0) and (4,3).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 4$.

Final answer

$x = 4$

Q052 **Written**

Find the equation of the line through (-1,2) and (-1,6).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -1$.

Final answer

$x = -1$

Worked Solutions **Triangles and Lines**

Q053 Written

Find the equation of the line through (-5,-3) and (-5,2).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -5$.

Final answer

$x = -5$

Q054 Written

Find the equation of the line through (-2,3) and (-3,3).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 3$.

Final answer

$y = 3$

Worked Solutions **Triangles and Lines**

Q055 **Written**

Find the equation of the line through (4,-2) and (1,-2).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -2$.

Final answer

$y = -2$

Q056 **Written**

Find the equation of the line through (-3,6) and (2,6).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 6$.

Final answer

$y = 6$

Worked Solutions **Triangles and Lines**

Q057 **Written**

Find the equation of the line through (1,2) and (3,10).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 4x - 2$.

Final answer

$y = 4x - 2$

Q058 **Written**

Find the equation of the line through (-1,-8) and (4,2).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 2x - 6$.

Final answer

$y = 2x - 6$

Q059 MCQ

Find the equation of the line through (-3,1) and (-3,2):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -3$.

Correct option: B

Final answer

$x = -3$

Q060 MCQ

Find the equation of the line through (-2,3) and (-5,3):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3$.

Correct option: C

Final answer

$y = 3$

Worked Solutions **Triangles and Lines**

Q061 MCQ

Find the equation of the line through (1,-1) and (4,0):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = x/3 - 4/3(\text{equivalently} x - 3y - 4 = 0)$.

Correct option: D

Final answer

$y = x/3 - 4/3(\text{equivalently} x - 3y - 4 = 0)$

Q062 MCQ

Find the equation of the line through (-2,-2) and (2,-2):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -2$.

Correct option: A

Final answer

$y = -2$

Worked Solutions **Triangles and Lines**

Q063 MCQ

Find the equation of the line through (-4,2) and (-4,8):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -4$.

Correct option: B

Final answer

$x = -4$

Q064 MCQ

Find the equation of the line through (-2,-6) and (-4,1):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $7x + 2y + 26 = 0$.

Correct option: D

Final answer

$7x + 2y + 26 = 0$

Worked Solutions **Triangles and Lines**

Q065 MCQ

Find the equation of the line through (4,0) and (4,3):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 4$.

Correct option: C

Final answer

$x = 4$

Q066 MCQ

Find the equation of the line through (-1,2) and (-1,6):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -1$.

Correct option: A

Final answer

$x = -1$

Worked Solutions **Triangles and Lines**

Q067 MCQ

Find the equation of the line through (-5,-3) and (-5,2):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -5$.

Correct option: C

Final answer

$x = -5$

Q068 MCQ

Find the equation of the line through (-2,3) and (-3,3):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3$.

Correct option: D

Final answer

$y = 3$

Worked Solutions **Triangles and Lines**

Q069 MCQ

Find the equation of the line through (4,-2) and (1,-2):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -2$.

Correct option: D

Final answer

$y = -2$

Q070 MCQ

Find the equation of the line through (-3,6) and (2,6):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 6$.

Correct option: C

Final answer

$y = 6$

Worked Solutions **Triangles and Lines**

Q071 MCQ

Find the equation of the line through (1,2) and (3,10):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 4x - 2$.

Correct option: D

Final answer

$y = 4x - 2$

Q072 MCQ

Find the equation of the line through (-1,-8) and (4,2):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 2x - 6$.

Correct option: D

Final answer

$y = 2x - 6$

Worked Solutions **Triangles and Lines**

Q073 **Written**

Find the equation of the line through (2,-1) with gradient 3.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 3x - 7$.

Final answer

$y = 3x - 7$

Q074 **Written**

Find the equation of the line through (-3,4) with gradient 2.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 2x + 10$.

Final answer

$y = 2x + 10$

Worked Solutions **Triangles and Lines**

Q075 Written

Find the equation of the line through (-1,5) with gradient -1.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -x + 4$.

Final answer

$y = -x + 4$

Q076 Written

Find the equation of the line through (-3,-8) with gradient 1.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = x - 5$.

Final answer

$y = x - 5$

Worked Solutions **Triangles and Lines**

Q077 Written

Find the equation of the line through (2,-1) with gradient -2.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -2x + 3$.

Final answer

$y = -2x + 3$

Q078 MCQ

Find the equation of the line through (2,-1) with gradient 3:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3x - 7$.

Correct option: D

Final answer

$y = 3x - 7$

Worked Solutions **Triangles and Lines**

Q079 MCQ

Find the equation of the line through (-3,4) with gradient 2:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 2x + 10$.

Correct option: C

Final answer

$y = 2x + 10$

Q080 MCQ

Find the equation of the line through (-1,5) with gradient -1:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -x + 4$.

Correct option: D

Final answer

$y = -x + 4$

Q081 MCQ

Find the equation of the line through (-3,-8) with gradient 1:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = x - 5$.

Correct option: B

Final answer

$y = x - 5$

Q082 MCQ

Find the equation of the line through (2,-1) with gradient -2:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -2x + 3$.

Correct option: D

Final answer

$y = -2x + 3$

Worked Solutions **Triangles and Lines**

Q083 **Challenge**

Sensors A $(-4,\ 2)$ and B $(9,\ 2)$ have the same height. Find the straight-line boundary through A and B.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $y = 2$.

Final answer

$y = 2$

Worked Solutions **Triangles and Lines**

Q084 **Challenge**

Sensors A $(-4,\ 2)$ and B $(9,\ 2)$ have the same height. Find the straight-line boundary through A and B:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 2$.

Correct option: D

Final answer

$y = 2$

Worked Solutions **Triangles and Lines**

Q085 Written

A $(2, \ \ 3)$, B $(4, \ \ -1)$, and C $(a-1, \ \ 2-a)$ are collinear. Find a.

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = 7$.

Final answer

$a = 7$

Q086 Written

A $(2, \ \ 5)$, B $(4, \ \ 9)$, and C $(-1, \ \ a)$ are collinear. Find a.

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = -1$.

Final answer

$a = -1$

Worked Solutions **Triangles and Lines**

Q087 Written

A $(1,\ \ -1)$, B $(-2,\ \ -3)$, and C $(4,\ \ a)$ are collinear. Find a.

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = 1$.

Final answer

$a = 1$

Q088 Written

A $(1,\ \ 2)$, B $(3,\ \ 1)$, and C $(a,\ \ -1)$ are collinear. Find a.

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = 7$.

Final answer

$a = 7$

Worked Solutions **Triangles and Lines**

Q089 Written

A $(-3, \quad -9)$, B $(5, \quad 7)$, and C $(a+3, \quad 3a-2)$ are collinear. Find a.

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = 5$.

Final answer

$a = 5$

Q090 MCQ

A $(2, \quad 3)$, B $(4, \quad -1)$, and C $(a-1, \quad 2-a)$ are collinear. Find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 7$.

Correct option: D

Final answer

$a = 7$

Worked Solutions **Triangles and Lines**

Q091 MCQ

A $(2,\quad 5)$, B $(4,\quad 9)$, and C $(-1,\quad a)$ are collinear. Find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = -1$.

Correct option: D

Final answer

$a = -1$

Q092 MCQ

A $(1,\quad -1)$, B $(-2,\quad -3)$, and C $(4,\quad a)$ are collinear. Find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 1$.

Correct option: C

Final answer

$a = 1$

Worked Solutions **Triangles and Lines**

Q093 MCQ

A $(1,\ \ 2)$, B $(3,\ \ 1)$, and C $(a,\ \ -1)$ are collinear. Find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 7$.

Correct option: D

Final answer

$a = 7$

Q094 MCQ

A $(-3,\ \ -9)$, B $(5,\ \ 7)$, and C $(a+3,\ \ 3a-2)$ are collinear. Find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 5$.

Correct option: D

Final answer

$a = 5$

Worked Solutions **Triangles and Lines**

Q095 Written

Find the line through the intersection of $2x - 3y + 1 = 0$ and $x + 2y - 3 = 0$, passing through (-1,3).

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $x + y - 2 = 0$.

Final answer

$x + y - 2 = 0$

Q096 Written

Find the line through the intersection of $x + 2y - 10 = 0$ and $2x + 3y - 7 = 0$, passing through (5,6).

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $x + 3y - 23 = 0$.

Final answer

$x + 3y - 23 = 0$

Worked Solutions **Triangles and Lines**

Q097 **Written**

Find the line through the intersection of $3x + 5y - 1 = 0$ and $7x + 8y - 6 = 0$, passing through (-1,2).

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $x + y - 1 = 0$.

Final answer

$x + y - 1 = 0$

Q098 **Written**

Find the line through the intersection of $x + 2y - 4 = 0$ and $2x - y - 3 = 0$, passing through (-3,-1).

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $2x - 5y + 1 = 0$.

Final answer

$2x - 5y + 1 = 0$

Worked Solutions **Triangles and Lines**

Q099 MCQ

Find the line through the intersection of $2x - 3y + 1 = 0$ and $x + 2y - 3 = 0$, passing through (-1,3):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + y - 2 = 0$.

Correct option: B

Final answer

$x + y - 2 = 0$

Q100 MCQ

Find the line through the intersection of $x + 2y - 10 = 0$ and $2x + 3y - 7 = 0$, passing through (5,6):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + 3y - 23 = 0$.

Correct option: A

Final answer

$x + 3y - 23 = 0$

Worked Solutions **Triangles and Lines**

Q101 *MCQ*

Find the line through the intersection of $3x + 5y - 1 = 0$ and $7x + 8y - 6 = 0$, passing through (-1,2):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + y - 1 = 0$.

Correct option: B

Final answer

$x + y - 1 = 0$

Q102 *MCQ*

Find the line through the intersection of $x + 2y - 4 = 0$ and $2x - y - 3 = 0$, passing through (-3,-1):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - 5y + 1 = 0$.

Correct option: D

Final answer

$2x - 5y + 1 = 0$

Worked Solutions **Triangles and Lines**

Q103 Challenge

The intersection of $6x + 5y + 7 = 0$ and $3x - 2y + 5 = 0$ is joined to the origin. Find the equation of that trajectory.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $3x + 13y = 0$.

Final answer

$3x + 13y = 0$

Worked Solutions **Triangles and Lines**

Q104 **Challenge**

The intersection of $6x + 5y + 7 = 0$ and $3x - 2y + 5 = 0$ is joined to the origin. Find the equation of that trajectory:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x + 13y = 0$.

Correct option: C

Final answer

$3x + 13y = 0$

Worked Solutions **Triangles and Lines**

Q105 Written

Find the gradient of line $y = 2x + 3$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 2$.

Final answer

$m = 2$

Q106 Written

Find the gradient of line $y = -3x + 2$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -3$.

Final answer

$m = -3$

Worked Solutions **Triangles and Lines**

Q107 **Written**

Find the gradient of line $x - 3y + 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 1/3$.

Final answer

$m = 1/3$

Q108 **Written**

Find the gradient of line $2x - y - 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 2$.

Final answer

$m = 2$

Worked Solutions **Triangles and Lines**

Q109 **Written**

Find the gradient of line $x + 3y + 3 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/3$.

Final answer

$m = -1/3$

Q110 **Written**

Find the gradient of line $y = 3x + 1$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3$.

Final answer

$m = 3$

Worked Solutions **Triangles and Lines**

Q111 Written

Find the gradient of line $y = 4x - 1$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 4$.

Final answer

$m = 4$

Q112 Written

Find the gradient of line $y = -4x - 7$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -4$.

Final answer

$m = -4$

Worked Solutions **Triangles and Lines**

Q113 Written

Find the gradient of line $6x - 2y = 5$**.**

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3$.

Final answer

$m = 3$

Q114 Written

Find the gradient of line $8x + 2y - 3 = 0$**.**

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -4$.

Final answer

$m = -4$

Worked Solutions **Triangles and Lines**

Q115 **Written**

Find the gradient of line $x + 4y - 4 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/4$.

Final answer

$m = -1/4$

Q116 **Written**

Find the gradient of line $y = 3x - 1$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3$.

Final answer

$m = 3$

Worked Solutions **Triangles and Lines**

Q117 Written

Find the gradient of line $y = -3x + 4$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -3$.

Final answer

$m = -3$

Q118 Written

Find the gradient of line $y = x - 3$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 1$.

Final answer

$m = 1$

Worked Solutions **Triangles and Lines**

Q119 Written

Find the gradient of line $6x + 2y + 5 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -3$.

Final answer

$m = -3$

Q120 Written

Find the gradient of line $2x - 2y - 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 1$.

Final answer

$m = 1$

Worked Solutions **Triangles and Lines**

Q121 **Written**

Find the gradient of line $x + 3y - 2 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/3$.

Final answer

$m = -1/3$

Q122 **Written**

Find the gradient of line $y = 2x - 1$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 2$.

Final answer

$m = 2$

Worked Solutions **Triangles and Lines**

Q123 **Written**

Find the gradient of line $y = -x + 4$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1$.

Final answer

$m = -1$

Q124 **Written**

Find the gradient of line $y = 3x/2 + 6$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3/2$.

Final answer

$m = 3/2$

Worked Solutions **Triangles and Lines**

Q125 Written

Find the gradient of line $x + y + 3 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1$.

Final answer

$m = -1$

Q126 Written

Find the gradient of line $2x + 3y + 8 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -2/3$.

Final answer

$m = -2/3$

Worked Solutions **Triangles and Lines**

Q127 Written

Find the gradient of line $3x + 6y - 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/2$.

Final answer

$m = -1/2$

Q128 MCQ

Find the gradient of line $y = 2x + 3$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 2$.

Correct option: D

Final answer

$m = 2$

Worked Solutions **Triangles and Lines**

Q129 MCQ

Find the gradient of line $y = -3x + 2$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -3$.

Correct option: C

Final answer

$m = -3$

Q130 MCQ

Find the gradient of line $x - 3y + 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 1/3$.

Correct option: A

Final answer

$m = 1/3$

Worked Solutions **Triangles and Lines**

Q131 MCQ

Find the gradient of line $2x - y - 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 2$.

Correct option: B

Final answer

$m = 2$

Q132 MCQ

Find the gradient of line $x + 3y + 3 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/3$.

Correct option: A

Final answer

$m = -1/3$

Worked Solutions **Triangles and Lines**

Q133 MCQ

Find the gradient of line $y = 3x + 1$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3$.

Correct option: A

Final answer

$m = 3$

Q134 MCQ

Find the gradient of line $y = 4x - 1$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 4$.

Correct option: B

Final answer

$m = 4$

Worked Solutions **Triangles and Lines**

Q135 MCQ

Find the gradient of line $y = -4x - 7$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -4$.

Correct option: D

Final answer

$m = -4$

Q136 MCQ

Find the gradient of line $6x - 2y = 5$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3$.

Correct option: A

Final answer

$m = 3$

Worked Solutions **Triangles and Lines**

Q137 MCQ

Find the gradient of line $8x + 2y - 3 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -4$.

Correct option: C

Final answer

$m = -4$

Q138 MCQ

Find the gradient of line $x + 4y - 4 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/4$.

Correct option: C

Final answer

$m = -1/4$

Worked Solutions **Triangles and Lines**

Q139 MCQ

Find the gradient of line $y = 3x - 1$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3$.

Correct option: D

Final answer

$m = 3$

Q140 MCQ

Find the gradient of line $y = -3x + 4$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -3$.

Correct option: A

Final answer

$m = -3$

Worked Solutions **Triangles and Lines**

Q141 MCQ

Find the gradient of line $y = x - 3$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 1$.

Correct option: D

Final answer

$m = 1$

Q142 MCQ

Find the gradient of line $6x + 2y + 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -3$.

Correct option: C

Final answer

$m = -3$

Worked Solutions **Triangles and Lines**

Q143 MCQ

Find the gradient of line $2x - 2y - 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 1$.

Correct option: C

Final answer

$m = 1$

Q144 MCQ

Find the gradient of line $x + 3y - 2 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/3$.

Correct option: A

Final answer

$m = -1/3$

Worked Solutions **Triangles and Lines**

Q145 MCQ

Find the gradient of line $y = 2x - 1$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 2$.

Correct option: D

Final answer

$m = 2$

Q146 MCQ

Find the gradient of line $y = -x + 4$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1$.

Correct option: A

Final answer

$m = -1$

Worked Solutions **Triangles and Lines**

Q147 MCQ

Find the gradient of line $y = 3x/2 + 6$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3/2$.

Correct option: B

Final answer

$m = 3/2$

Q148 MCQ

Find the gradient of line $x + y + 3 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1$.

Correct option: D

Final answer

$m = -1$

Worked Solutions **Triangles and Lines**

Q149 MCQ

Find the gradient of line $2x + 3y + 8 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -2/3$.

Correct option: B

Final answer

$m = -2/3$

Q150 MCQ

Find the gradient of line $3x + 6y - 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/2$.

Correct option: B

Final answer

$m = -1/2$

Worked Solutions **Triangles and Lines**

Q151 **Written**

Find the line through (2,-4) parallel to $3x - 2y + 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $3x - 2y - 14 = 0$.

Final answer

$3x - 2y - 14 = 0$

Q152 **Written**

Find the line through (1,-3) parallel to $6x + 3y - 5 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + y + 1 = 0$.

Final answer

$2x + y + 1 = 0$

Worked Solutions **Triangles and Lines**

Q153 **Written**

Find the equation of the line through (1,-2) parallel to $2x - y + 5 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x - y - 4 = 0$.

Final answer

$2x - y - 4 = 0$

Q154 **Written**

Find the equation of the line through (-1,2) parallel to $2x + 3y = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + 3y - 4 = 0$.

Final answer

$2x + 3y - 4 = 0$

Worked Solutions **Triangles and Lines**

Q155 Written

Find the equation of the line through (3,-1) parallel to $3x + 2y + 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $3x + 2y - 7 = 0$.

Final answer

$3x + 2y - 7 = 0$

Q156 MCQ

Find the line through (2,-4) parallel to $3x - 2y + 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x - 2y - 14 = 0$.

Correct option: C

Final answer

$3x - 2y - 14 = 0$

Worked Solutions **Triangles and Lines**

Q157 MCQ

Find the line through (1,-3) parallel to $6x + 3y - 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + y + 1 = 0$.

Correct option: D

Final answer

$2x + y + 1 = 0$

Q158 MCQ

Find the equation of the line through (1,-2) parallel to $2x - y + 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - y - 4 = 0$.

Correct option: C

Final answer

$2x - y - 4 = 0$

Worked Solutions | **Triangles and Lines**

Q159 MCQ

Find the equation of the line through (-1,2) parallel to $2x + 3y = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + 3y - 4 = 0$.

Correct option: D

Final answer

$2x + 3y - 4 = 0$

Q160 MCQ

Find the equation of the line through (3,-1) parallel to $3x + 2y + 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x + 2y - 7 = 0$.

Correct option: D

Final answer

$3x + 2y - 7 = 0$

Worked Solutions **Triangles and Lines**

Q161 Written

Find the line through (-2,5) perpendicular to $3x + 5y + 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $5x - 3y + 25 = 0$.

Final answer

$5x - 3y + 25 = 0$

Q162 Written

Find the line through (2,1) perpendicular to $2x + 3y + 4 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $3x - 2y - 4 = 0$.

Final answer

$3x - 2y - 4 = 0$

Worked Solutions **Triangles and Lines**

Q163 **Written**

Find the equation of the line through (2,-5) perpendicular to $2x + y - 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $x - 2y - 12 = 0$.

Final answer

$x - 2y - 12 = 0$

Q164 **Written**

Find the equation of the line through (-1,3) perpendicular to $5x - 2y + 3 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + 5y - 13 = 0$.

Final answer

$2x + 5y - 13 = 0$

Worked Solutions **Triangles and Lines**

Q165 Written

Find the equation of the line through (-2,-5) perpendicular to $3x - 2y - 7 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + 3y + 19 = 0$.

Final answer

$2x + 3y + 19 = 0$

Q166 MCQ

Find the line through (-2,5) perpendicular to $3x + 5y + 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $5x - 3y + 25 = 0$.

Correct option: D

Final answer

$5x - 3y + 25 = 0$

Worked Solutions **Triangles and Lines**

Q167 MCQ

Find the line through (2,1) perpendicular to $2x + 3y + 4 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x - 2y - 4 = 0$.

Correct option: D

Final answer

$3x - 2y - 4 = 0$

Q168 MCQ

Find the equation of the line through (2,-5) perpendicular to $2x + y - 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x - 2y - 12 = 0$.

Correct option: B

Final answer

$x - 2y - 12 = 0$

Worked Solutions **Triangles and Lines**

Q169 MCQ

Find the equation of the line through (-1,3) perpendicular to $5x - 2y + 3 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + 5y - 13 = 0$.

Correct option: A

Final answer

$2x + 5y - 13 = 0$

Q170 MCQ

Find the equation of the line through (-2,-5) perpendicular to $3x - 2y - 7 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + 3y + 19 = 0$.

Correct option: D

Final answer

$2x + 3y + 19 = 0$

Worked Solutions **Triangles and Lines**

Q171 Challenge

If $L1 : 2x + 3y - 10 = 0$ is perpendicular to $L2 : kx - 2y + 5 = 0$ and parallel to $L3 : 4x + my - 1 = 0$, find m/k.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $m/k = 2\,(k = 3,\quad m = 6)$.

Final answer

$m/k = 2\,(k = 3,\quad m = 6)$

Worked Solutions **Triangles and Lines**

Q172 **Challenge**

If $L1 : 2x + 3y - 10 = 0$ is perpendicular to $L2 : kx - 2y + 5 = 0$ and parallel to $L3 : 4x + my - 1 = 0$, find m/k:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m/k = 2\,(k = 3,\quad m = 6)$.

Correct option: D

Final answer

$m/k = 2\,(k = 3,\quad m = 6)$

Worked Solutions **Triangles and Lines**

Q173 **Written**

For A $(5,\ \ 11)$, B $(1,\ \ -1)$, find the equation of the perpendicular bisector of AB.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $x + 3y - 18 = 0$.

Final answer

$x + 3y - 18 = 0$

Q174 **Written**

For A $(-1,\ \ 2)$, B $(3,\ \ -4)$, find the equation of the perpendicular bisector of AB.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $2x - 3y - 5 = 0$.

Final answer

$2x - 3y - 5 = 0$

Worked Solutions **Triangles and Lines**

Q175 Written

Find the perpendicular bisector of segment A $(-2,\ 3)$, B $(2,\ 1)$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $2x - y + 2 = 0$.

Final answer

$2x - y + 2 = 0$

Q176 Written

Find the perpendicular bisector of segment A $(-1,\ 2)$, B $(5,\ -2)$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $3x - 2y - 6 = 0$.

Final answer

$3x - 2y - 6 = 0$

Worked Solutions **Triangles and Lines**

Q177 Written

Find the perpendicular bisector of segment A $(0,\ \ 6)$, B $(4,\ \ 4)$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $2x - y + 1 = 0$.

Final answer

$2x - y + 1 = 0$

Q178 MCQ

For A $(5,\ \ 11)$, B $(1,\ \ -1)$, find the equation of the perpendicular bisector of AB:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + 3y - 18 = 0$.

Correct option: A

Final answer

$x + 3y - 18 = 0$

Worked Solutions **Triangles and Lines**

Q179 MCQ

For A $(-1,\ 2)$, B $(3,\ -4)$, find the equation of the perpendicular bisector of AB:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - 3y - 5 = 0$.

Correct option: C

Final answer

$2x - 3y - 5 = 0$

Q180 MCQ

Find the perpendicular bisector of segment A $(-2,\ 3)$, B $(2,\ 1)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - y + 2 = 0$.

Correct option: C

Final answer

$2x - y + 2 = 0$

Worked Solutions **Triangles and Lines**

Q181 *MCQ*

Find the perpendicular bisector of segment A $(-1,\ \ 2)$, B $(5,\ \ -2)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x - 2y - 6 = 0$.

Correct option: D

Final answer

$3x - 2y - 6 = 0$

Q182 *MCQ*

Find the perpendicular bisector of segment A $(0,\ \ 6)$, B $(4,\ \ 4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - y + 1 = 0$.

Correct option: A

Final answer

$2x - y + 1 = 0$

Worked Solutions **Triangles and Lines**

Q183 Written

Find the point B symmetric to A $(-4,\ \ 3)$ with respect to $3x - y + 5 = 0$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (2,\ \ 1)$.

Final answer

$B = (2,\ \ 1)$

Q184 Written

Find the point B symmetric to A $(0,\ \ 4)$ with respect to $2x - y - 1 = 0$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (4,\ \ 2)$.

Final answer

$B = (4,\ \ 2)$

Worked Solutions **Triangles and Lines**

Q185 **Written**

Find the point B symmetric to A $(1,\ \ 2)$ with respect to $x + 2y - 10 = 0$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (3,\ \ 6)$.

Final answer

$B = (3,\ \ 6)$

Q186 **Written**

Find the point B symmetric to A $(4,\ \ -1)$ with respect to $4x - 2y - 3 = 0$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (-2,\ \ 2)$.

Final answer

$B = (-2,\ \ 2)$

Worked Solutions **Triangles and Lines**

Q187 MCQ

Find the point B symmetric to A $(-4,\ \ 3)$ with respect to $3x - y + 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (2,\ \ 1)$.

Correct option: A

Final answer

$B = (2,\ \ 1)$

Q188 MCQ

Find the point B symmetric to A $(0,\ \ 4)$ with respect to $2x - y - 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (4,\ \ 2)$.

Correct option: A

Final answer

$B = (4,\ \ 2)$

Worked Solutions **Triangles and Lines**

Q189 MCQ

Find the point B symmetric to A $(1,\quad 2)$ with respect to $x + 2y - 10 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (3,\quad 6)$.

Correct option: C

Final answer

$B = (3,\quad 6)$

Q190 MCQ

Find the point B symmetric to A $(4,\quad -1)$ with respect to $4x - 2y - 3 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (-2,\quad 2)$.

Correct option: C

Final answer

$B = (-2,\quad 2)$

Worked Solutions **Triangles and Lines**

Q191 Challenge

Reflect A $(5,\quad 0)$ across the targeting line $y = 2x$. Find the receiver point B.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $B = (-3,\quad 4)$.

Final answer

$B = (-3,\quad 4)$

Worked Solutions **Triangles and Lines**

Q192 **Challenge**

Reflect A $(5,\ \ 0)$ across the targeting line $y = 2x$. Find the receiver point B:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (-3,\ \ 4)$.

Correct option: C

Final answer

$B = (-3,\ \ 4)$

Worked Solutions **Triangles and Lines**

Q193 **Written**

Find the distance from (-2,1) to $3x - 2y + 5 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $3/\sqrt{13} = 3\sqrt{13}/13$.

Final answer

$3/\sqrt{13} = 3\sqrt{13}/13$

Q194 **Written**

Find the distance from (3,-2) to $x + 2y - 4 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{5}$.

Final answer

$\sqrt{5}$

Worked Solutions **Triangles and Lines**

Q195 Written

Find the perpendicular distance from (-3,2) to $2x - 3y + 6 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $6/\sqrt{13} = 6\sqrt{13}/13$.

Final answer

$6/\sqrt{13} = 6\sqrt{13}/13$

Q196 Written

Find the perpendicular distance from (-3,7) to $12x - 5y = 7$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: 6.

Final answer

6

Worked Solutions **Triangles and Lines**

Q197 **Written**

Find the perpendicular distance from (-8,1) to $y = 2x - 3$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $4\sqrt{5}$.

Final answer

$4\sqrt{5}$

Q198 **Written**

Find the perpendicular distance from the origin to $3x - 4y - 5 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: 1.

Final answer

1

Worked Solutions **Triangles and Lines**

Q199 Written

Find the perpendicular distance from (2,3) to $3x + 2y + 1 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{13}$.

Final answer

$\sqrt{13}$

Q200 Written

Find the perpendicular distance from (13,4) to $3x + 4y - 5 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: 10.

Final answer

10

Worked Solutions **Triangles and Lines**

Q201 Written

Find the perpendicular distance from (1,2) to $y = 3x + 1$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{10}/5$.

Final answer

$\sqrt{10}/5$

Q202 MCQ

Find the distance from (-2,1) to $3x - 2y + 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3/\sqrt{13} = 3\sqrt{13}/13$.

Correct option: A

Final answer

$3/\sqrt{13} = 3\sqrt{13}/13$

Worked Solutions **Triangles and Lines**

Q203 MCQ

Find the distance from (3,-2) to $x + 2y - 4 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{5}$.

Correct option: D

Final answer

$\sqrt{5}$

Q204 MCQ

Find the perpendicular distance from (-3,2) to $2x - 3y + 6 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $6/\sqrt{13} = 6\sqrt{13}/13$.

Correct option: D

Final answer

$6/\sqrt{13} = 6\sqrt{13}/13$

Worked Solutions **Triangles and Lines**

Q205 MCQ

Find the perpendicular distance from (-3,7) to $12x - 5y = 7$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 6.

Correct option: B

Final answer

6

Q206 MCQ

Find the perpendicular distance from (-8,1) to $y = 2x - 3$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $4\sqrt{5}$.

Correct option: D

Final answer

$4\sqrt{5}$

Worked Solutions **Triangles and Lines**

Q207 MCQ

Find the perpendicular distance from the origin to $3x - 4y - 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 1.

Correct option: D

Final answer

1

Q208 MCQ

Find the perpendicular distance from (2,3) to $3x + 2y + 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{13}$.

Correct option: A

Final answer

$\sqrt{13}$

Worked Solutions | **Triangles and Lines**

Q209 MCQ

Find the perpendicular distance from (13,4) to $3x + 4y - 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 10.

Correct option: B

Final answer

10

Q210 MCQ

Find the perpendicular distance from (1,2) to $y = 3x + 1$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}/5$.

Correct option: D

Final answer

$\sqrt{10}/5$

Worked Solutions **Triangles and Lines**

Q211 Written

Find the perpendicular length from the origin to $3x - y = 5$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{10}/2$.

Final answer
$\sqrt{10}/2$

Q212 Written

Find the perpendicular length from (-1,5) to $y = 3x - 2$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{10}$.

Final answer
$\sqrt{10}$

Worked Solutions **Triangles and Lines**

Q213 MCQ

Find the perpendicular length from the origin to $3x - y = 5$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}/2$.

Correct option: D

Final answer

$\sqrt{10}/2$

Q214 MCQ

Find the perpendicular length from (-1,5) to $y = 3x - 2$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}$.

Correct option: B

Final answer

$\sqrt{10}$

Worked Solutions **Triangles and Lines**

Q215 Challenge

Find the shortest distance from P $(-1,\ \ 5)$ to the radiation line $y = x/3 + 2$.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $\sqrt{10}$.

Final answer

$\sqrt{10}$

Worked Solutions **Triangles and Lines**

Q216 Challenge

Find the shortest distance from P $(-1,\ \ 5)$ to the radiation line $y = x/3 + 2$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}$.

Correct option: A

Final answer

$\sqrt{10}$

Worked Solutions **Triangles and Lines**

Q217 **Written**

Find the area of triangle ABC with A $(1,\quad 1)$, B $(3,\quad 7)$, C $(-3,\quad -1)$.

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 10.

Final answer

10

Q218 **Written**

Find the area for A $(1,\quad 5)$, B $(-3,\quad -3)$, C $(3,\quad -6)$.

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 30.

Final answer

30

Worked Solutions **Triangles and Lines**

Q219 **Written**

Find the area for A $(-2,\ \ 1)$**, B** $(2,\ \ -1)$**, C** $(0,\ \ 4)$**.**

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 8.

Final answer

8

Q220 **Written**

Find the area for A $(0,\ \ -6)$**, B** $(2,\ \ -2)$**, C** $(1,\ \ 5)$**.**

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 9.

Final answer

9

Worked Solutions **Triangles and Lines**

Q221 Written

Find the area for A $(-2,\ 4)$, B $(-3,\ -5)$, C $(5,\ -1)$.

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 34.

Final answer

34

Q222 Written

Find the area for A $(2,\ 5)$, B $(0,\ 1)$, C $(4,\ 2)$.

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 7.

Final answer

7

Worked Solutions **Triangles and Lines**

Q223 MCQ

Find the area of triangle ABC with A $(1, \ 1)$, B $(3, \ 7)$, C $(-3, \ -1)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 10.

Correct option: D

Final answer

10

Q224 MCQ

Find the area for A $(1, \ 5)$, B $(-3, \ -3)$, C $(3, \ -6)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 30.

Correct option: C

Final answer

30

Worked Solutions **Triangles and Lines**

Q225 MCQ

Find the area for A $(-2,\ \ 1)$, B $(2,\ \ -1)$, C $(0,\ \ 4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 8.

Correct option: A

Final answer

8

Q226 MCQ

Find the area for A $(0,\ \ -6)$, B $(2,\ \ -2)$, C $(1,\ \ 5)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 9.

Correct option: B

Final answer

9

Worked Solutions **Triangles and Lines**

Q227 MCQ

Find the area for A $(-2, \ 4)$, **B** $(-3, \ -5)$, **C** $(5, \ -1)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 34.

Correct option: B

Final answer

34

Q228 MCQ

Find the area for A $(2, \ 5)$, **B** $(0, \ 1)$, **C** $(4, \ 2)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 7.

Correct option: D

Final answer

7

Q229 Challenge

Find the area of the wound triangle A $(-2, \quad 1)$, B $(1, \quad 5)$, C $(-21, \quad -21)$.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: 5.

Final answer

5

Worked Solutions **Triangles and Lines**

Q230 Challenge

Find the area of the wound triangle A $(-2,\ \ 1)$, B $(1,\ \ 5)$, C $(-21,\ \ -21)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 5.

Correct option: D

Final answer

5

Worked Solutions **Triangles and Lines**

Q231 Written

The line $(k-4)x+(k+2)y+k+5=0$ passes through a fixed point for every k. Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(1/2,\quad -3/2)$.

Final answer

$(1/2,\quad -3/2)$

Q232 Written

The line $(2k+1)x-(k+2)y+7k+8=0$ passes through a fixed point for every k. Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(-2,\quad 3)$.

Final answer

$(-2,\quad 3)$

Worked Solutions **Triangles and Lines**

Q233 **Written**

The line $(k+2)x + ky - 6 = 0$ passes through a fixed point for every k. Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(3, \quad -3)$.

Final answer

$(3, \quad -3)$

Q234 **Written**

The line $(k+3)x - (2k+1)y + k - 2 = 0$ passes through a fixed point for every k. Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(1, \quad 1)$.

Final answer

$(1, \quad 1)$

Worked Solutions **Triangles and Lines**

Q235 Written

The line $(k+3)x+(3k+2)y+7=0$ passes through a fixed point for every k. Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(-3,\ 1)$.

Final answer

$(-3,\ 1)$

Q236 MCQ

The line $(k-4)x+(k+2)y+k+5=0$ passes through a fixed point for every k. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(1/2,\ -3/2)$.

Correct option: D

Final answer

$(1/2,\ -3/2)$

Worked Solutions **Triangles and Lines**

Q237 MCQ

The line $(2k+1)x-(k+2)y+7k+8=0$ passes through a fixed point for every k. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(-2,\ \ 3)$.

Correct option: B

Final answer

$(-2,\ \ 3)$

Q238 MCQ

The line $(k+2)x+ky-6=0$ passes through a fixed point for every k. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(3,\ \ -3)$.

Correct option: D

Final answer

$(3,\ \ -3)$

Worked Solutions Triangles and Lines

Q239 MCQ

The line $(k+3)x-(2k+1)y+k-2=0$ passes through a fixed point for every k. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(1,\quad 1)$.

Correct option: D

Final answer

$(1,\quad 1)$

Q240 MCQ

The line $(k+3)x+(3k+2)y+7=0$ passes through a fixed point for every k. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(-3,\quad 1)$.

Correct option: A

Final answer

$(-3,\quad 1)$

Worked Solutions **Triangles and Lines**

Q241 Challenge

The line $-(k-2)x + (k-3)y = -3k + 5$ passes through a fixed point for every k. Find it.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $(4,\quad 1)$.

Final answer

$(4,\quad 1)$

Worked Solutions **Triangles and Lines**

Q242 Challenge

The line $-(k-2)x+(k-3)y=-3k+5$ passes through a fixed point for every k. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(4,\quad 1)$.

Correct option: A

Final answer

$(4,\quad 1)$

Worked Solutions **Triangles and Lines**

Q243 Written

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a.

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = -4,\ -3,\ or 6$.

Final answer

$a = -4,\ -3,\ or 6$

Q244 Written

For $x - 2y + 8 = 0$, $x + y - 1 = 0$, and $ax + y - 5 = 0$ not to form a triangle, find a.

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = -1,\ -1/2,\ or 1$.

Final answer

$a = -1,\ -1/2,\ or 1$

Worked Solutions **Triangles and Lines**

Q245 Written

For $x - 2y + 1 = 0$, $3x + 2y = 6$, and $ax - 3y + 2 = 0$ not to form a triangle, find a.

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = 11/10,\ 3/2,\ or - 9/2$.

Final answer

$a = 11/10,\ 3/2,\ or - 9/2$

Q246 Written

For $x - y = -1$, $3x + 2y = 12$, and $ax - y = a - 1$ not to form a triangle, find a.

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = 1,\ -3/2,\ or 2$.

Final answer

$a = 1,\ -3/2,\ or 2$

Worked Solutions **Triangles and Lines**

Q247 MCQ

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = -4,\ -3,\ or 6$.

Correct option: A

Final answer

$a = -4,\ -3,\ or 6$

Q248 MCQ

For $x - 2y + 8 = 0$, $x + y - 1 = 0$, and $ax + y - 5 = 0$ not to form a triangle, find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = -1,\ -1/2,\ or 1$.

Correct option: A

Final answer

$a = -1,\ -1/2,\ or 1$

Worked Solutions Triangles and Lines

Q249 MCQ

For $x - 2y + 1 = 0$, $3x + 2y = 6$, and $ax - 3y + 2 = 0$ not to form a triangle, find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 11/10,\ 3/2,\ or - 9/2$.

Correct option: C

Final answer

$a = 11/10,\ 3/2,\ or - 9/2$

Q250 MCQ

For $x - y = -1$, $3x + 2y = 12$, and $ax - y = a - 1$ not to form a triangle, find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 1,\ -3/2,\ or2$.

Correct option: D

Final answer

$a = 1,\ -3/2,\ or2$

Worked Solutions **Triangles and Lines**

Q251 **Challenge**

For $x + 2y - 1 = 0$, $3x + y + 2 = 0$, and $mx + 4y - 3 = 0$ not to form a triangle, find the sum of all possible m.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: 15.

Final answer

15

Worked Solutions **Triangles and Lines**

Q252 **Challenge**

For $x + 2y - 1 = 0$, $3x + y + 2 = 0$, and $mx + 4y - 3 = 0$ not to form a triangle, find the sum of all possible m:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 15.

Correct option: A

Final answer

15

Worked Solutions **Triangles and Lines**

Quiz MCQ 1 Concept Quiz · MCQ

Determine the shape of triangle A $(0,\ 0)$, B $(3,\ 0)$, C $(0,\ 4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $scalene\ right - angled$.

Correct option: C

> **Final answer**
> $scalene\ right - angled$

Quiz MCQ 2 Concept Quiz · MCQ

Find the equation of the line through (2,-1) with gradient 3:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3x - 7$.

Correct option: B

> **Final answer**
> $y = 3x - 7$

Worked Solutions **Triangles and Lines**

Quiz MCQ 3 Concept Quiz · MCQ

Find the perpendicular distance from (1,2) to $3x + 4y - 10 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 1/5.

Correct option: D

Final answer

1/5

Quiz MCQ 4 Concept Quiz · MCQ

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(3, \quad -3)$.

Correct option: C

Final answer

$(3, \quad -3)$

Worked Solutions **Triangles and Lines**

Quiz WRITTEN 5 Concept Quiz · Written

Find the area of triangle A $(0,\ 0)$, B $(4,\ 0)$, C $(0,\ 3)$.

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 6.

Final answer

6

Quiz WRITTEN 6 Concept Quiz · Written

Find the perpendicular bisector of the segment joining A $(1,\ 2)$ and B $(5,\ 4)$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $y = -2x + 9$.

Final answer

$y = -2x + 9$

Worked Solutions **Triangles and Lines**

Quiz WRITTEN 7 **Concept Quiz · Written**

Find the equation of the line through (1,2) perpendicular to $y = 3x - 1$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $y = -x/3 + 7/3$.

Final answer

$y = -x/3 + 7/3$

Quiz WRITTEN 8 **Concept Quiz · Written**

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a.

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = -4,\ -3,\ or 6$.

Final answer

$a = -4,\ -3,\ or 6$

Answer key

Use the question number shown on each page.

Q001 *equilateral*
Q002 *equilateral*
Q003 D
Q004 C
Q005 *isosceles right – angled; right angle at A*
Q006 *isosceles right – angled; right angle at A*
Q007 B
Q008 D
Q009 *scalene right – angled; right angle at A*
Q010 D
Q011 *isosceles right – angled; right angle at B*
Q012 *isosceles right – angled; right angle at B*
Q013 *isosceles right – angled; right angle at C*
Q014 B
Q015 C
Q016 B
Q017 $k = 2$
Q018 C
Q019 $y = -4$
Q020 $x = -3$
Q021 $y = 6$
Q022 $x = 3$
Q023 $y = 5$
Q024 $y = 1$
Q025 $y = -7$
Q026 $x = 1$
Q027 $x = -3$
Q028 $x = 3$
Q029 $x = 2$
Q030 $x = 4$
Q031 $x = -6$
Q032 B
Q033 C
Q034 D
Q035 D
Q036 B
Q037 D
Q038 B
Q039 C
Q040 D
Q041 D
Q042 B
Q043 D
Q044 A
Q045 $x = -3$
Q046 $y = 3$
Q047 $y = x/3 - 4/3(\text{equivalently} x - 3y - 4 = 0)$
Q048 $y = -2$
Q049 $x = -4$
Q050 $7x + 2y + 26 = 0$
Q051 $x = 4$
Q052 $x = -1$
Q053 $x = -5$
Q054 $y = 3$
Q055 $y = -2$
Q056 $y = 6$
Q057 $y = 4x - 2$
Q058 $y = 2x - 6$
Q059 B
Q060 C
Q061 D
Q062 A
Q063 B
Q064 D
Q065 C
Q066 A
Q067 C
Q068 D
Q069 D
Q070 C
Q071 D
Q072 D
Q073 $y = 3x - 7$
Q074 $y = 2x + 10$
Q075 $y = -x + 4$
Q076 $y = x - 5$
Q077 $y = -2x + 3$
Q078 D
Q079 C
Q080 D
Q081 B
Q082 D
Q083 $y = 2$
Q084 D
Q085 $a = 7$
Q086 $a = -1$
Q087 $a = 1$
Q088 $a = 7$
Q089 $a = 5$
Q090 D
Q091 D
Q092 C
Q093 D
Q094 D
Q095 $x + y - 2 = 0$
Q096 $x + 3y - 23 = 0$
Q097 $x + y - 1 = 0$
Q098 $2x - 5y + 1 = 0$
Q099 B
Q100 A
Q101 B
Q102 D
Q103 $3x + 13y = 0$
Q104 C

Q105 $m = 2$
Q106 $m = -3$
Q107 $m = 1/3$
Q108 $m = 2$
Q109 $m = -1/3$
Q110 $m = 3$
Q111 $m = 4$
Q112 $m = -4$
Q113 $m = 3$
Q114 $m = -4$
Q115 $m = -1/4$
Q116 $m = 3$
Q117 $m = -3$
Q118 $m = 1$
Q119 $m = -3$
Q120 $m = 1$
Q121 $m = -1/3$
Q122 $m = 2$
Q123 $m = -1$
Q124 $m = 3/2$
Q125 $m = -1$
Q126 $m = -2/3$
Q127 $m = -1/2$
Q128 D
Q129 C
Q130 A
Q131 B
Q132 A
Q133 A
Q134 B
Q135 D
Q136 A
Q137 C
Q138 C
Q139 D
Q140 A
Q141 D
Q142 C
Q143 C
Q144 A
Q145 D
Q146 A
Q147 B
Q148 D
Q149 B
Q150 B
Q151 $3x - 2y - 14 = 0$
Q152 $2x + y + 1 = 0$
Q153 $2x - y - 4 = 0$
Q154 $2x + 3y - 4 = 0$
Q155 $3x + 2y - 7 = 0$
Q156 C
Q157 D
Q158 C
Q159 D
Q160 D
Q161 $5x - 3y + 25 = 0$
Q162 $3x - 2y - 4 = 0$
Q163 $x - 2y - 12 = 0$
Q164 $2x + 5y - 13 = 0$
Q165 $2x + 3y + 19 = 0$
Q166 D
Q167 D
Q168 B
Q169 A
Q170 D
Q171 $m/k = 2\,(k = 3,\ \ m = 6)$
Q172 D
Q173 $x + 3y - 18 = 0$
Q174 $2x - 3y - 5 = 0$
Q175 $2x - y + 2 = 0$
Q176 $3x - 2y - 6 = 0$
Q177 $2x - y + 1 = 0$
Q178 A
Q179 C
Q180 C
Q181 D
Q182 A
Q183 $B = (2,\ \ 1)$
Q184 $B = (4,\ \ 2)$
Q185 $B = (3,\ \ 6)$
Q186 $B = (-2,\ \ 2)$
Q187 A
Q188 A
Q189 C
Q190 C
Q191 $B = (-3,\ \ 4)$
Q192 C
Q193 $3/\sqrt{13} = 3\sqrt{13}/13$
Q194 $\sqrt{5}$
Q195 $6/\sqrt{13} = 6\sqrt{13}/13$
Q196 6
Q197 $4\sqrt{5}$
Q198 1
Q199 $\sqrt{13}$
Q200 10
Q201 $\sqrt{10}/5$
Q202 A
Q203 D
Q204 D
Q205 B
Q206 D
Q207 D
Q208 A
Q209 B
Q210 D
Q211 $\sqrt{10}/2$
Q212 $\sqrt{10}$
Q213 D
Q214 B
Q215 $\sqrt{10}$
Q216 A
Q217 10
Q218 30
Q219 8
Q220 9
Q221 34
Q222 7
Q223 D
Q224 C
Q225 A
Q226 B
Q227 B
Q228 D
Q229 5
Q230 D
Q231 $(1/2,\ \ -3/2)$
Q232 $(-2,\ \ 3)$

Q233 $(3,\ -3)$

Q234 $(1,\ 1)$

Q235 $(-3,\ 1)$

Q236 D

Q237 B

Q238 D

Q239 D

Q240 A

Q241 $(4,\ 1)$

Q242 A

Q243 $a = -4,\ -3,\ or6$

Q244 $a = -1,\ -1/2,\ or1$

Q245 $a = 11/10,\ 3/2,\ or - 9/2$

Q246 $a = 1,\ -3/2,\ or2$

Q247 A

Q248 A

Q249 C

Q250 D

Q251 15

Q252 A

Quiz MCQ 1 C

Quiz MCQ 2 B

Quiz MCQ 3 D

Quiz MCQ 4 C

Quiz WRITTEN 5 6

Quiz WRITTEN 6 $y = -2x + 9$

Quiz WRITTEN 7 $y = -x/3 + 7/3$

Quiz WRITTEN 8 $a = -4,\ -3,\ or6$

Concept 3 | Circles

English Student Edition • Focused formulas and guided practice

Formula opener

Circle
$$(x-a)^2+(y-b)^2=r^2$$

General form
$$x^2+y^2+2gx+2fy+c=0$$

Tangent
$$xx_1+yy_1=r^2$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Circles**

Q001 **Written**

Find the equation of the circle with centre (3,-2) and radius 5.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2+(y-k)^2=r^2$ and compare coefficients when needed.
3. Circle: $(x-3)^2+(y+2)^2=25$.

Final answer

$(x-3)^2+(y+2)^2=25$

Q002 **Written**

Find the equation of the circle with centre (-3,1) and radius 2.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2+(y-k)^2=r^2$ and compare coefficients when needed.
3. Circle: $(x+3)^2+(y-1)^2=4$.

Final answer

$(x+3)^2+(y-1)^2=4$

Worked Solutions **Circles**

Q003 Written

Find the equation of the circle with centre the origin and radius 4.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2+(y-k)^2=r^2$ and compare coefficients when needed.
3. Circle: $x^2+y^2=16$.

Final answer

$x^2+y^2=16$

Q004 Written

Find the equation of the circle with centre (3,4), radius 6.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2+(y-k)^2=r^2$ and compare coefficients when needed.
3. Circle: $(x-3)^2+(y-4)^2=36$.

Final answer

$(x-3)^2+(y-4)^2=36$

Worked Solutions **Circles**

Q005 **Written**

Find the equation of the circle with centre (-2,6), radius 2sqrt(3).

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2+(y-k)^2=r^2$ and compare coefficients when needed.
3. Circle: $(x+2)^2+(y-6)^2=12$.

Final answer

$(x+2)^2+(y-6)^2=12$

Q006 **Written**

Find the equation of the circle with centre the origin, radius 5.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2+(y-k)^2=r^2$ and compare coefficients when needed.
3. Circle: $x^2+y^2=25$.

Final answer

$x^2+y^2=25$

Worked Solutions **Circles**

Q007 MCQ

Find the equation of the circle with centre (3,-2) and radius 5:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-3)^2 + (y+2)^2 = 25$.

Correct option: A

Final answer

$(x-3)^2 + (y+2)^2 = 25$

Q008 MCQ

Find the equation of the circle with centre (-3,1) and radius 2:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+3)^2 + (y-1)^2 = 4$.

Correct option: A

Final answer

$(x+3)^2 + (y-1)^2 = 4$

Worked Solutions **Circles**

Q009 MCQ

Find the equation of the circle with centre the origin and radius 4:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 = 16$.

Correct option: D

Final answer

$x^2 + y^2 = 16$

Q010 MCQ

Find the equation of the circle with centre (3,4), radius 6:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-3)^2 + (y-4)^2 = 36$.

Correct option: B

Final answer

$(x-3)^2 + (y-4)^2 = 36$

Worked Solutions **Circles**

Q011 MCQ

Find the equation of the circle with centre (-2,6), radius 2sqrt(3):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+2)^2+(y-6)^2=12$.

Correct option: B

Final answer

$(x+2)^2+(y-6)^2=12$

Q012 MCQ

Find the equation of the circle with centre the origin, radius 5:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2+y^2=25$.

Correct option: A

Final answer

$x^2+y^2=25$

Worked Solutions **Circles**

Q013 **Written**

Determine what kind of figure $x^2 + y^2 - 6x + 2y + 6 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2 + (y-k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(3, -1)$, radius 2.

Final answer

circle; centre $(3, -1)$, radius 2

Q014 **Written**

Determine what kind of figure $x^2 + y^2 - 4x - 6y + 4 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2 + (y-k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(2, 3)$, radius 3.

Final answer

circle; centre $(2, 3)$, radius 3

Worked Solutions **Circles**

Q015 Written

Determine what kind of figure $x^2 + y^2 - 4x + 8y + 4 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2 + (y-k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle ; centre $(2, -4)$, radius 4.

Final answer

circle ; centre $(2, -4)$, radius 4

Q016 Written

Determine what kind of figure $x^2 + y^2 - 4x = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2 + (y-k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle ; centre $(2, 0)$, radius 2.

Final answer

circle ; centre $(2, 0)$, radius 2

Worked Solutions **Circles**

Q017 Written

Determine what kind of figure $x^2 + y^2 + 2x - 4y - 20 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $\text{circle}\,;\ \text{centre}\ (-1,\ 2)\,,\ \text{radius}\,5$.

Final answer

$\text{circle}\,;\ \text{centre}\ (-1,\ 2)\,,\ \text{radius}\,5$

Q018 Written

Determine what kind of figure $x^2 + y^2 + 4x - 6y = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $\text{circle}\,;\ \text{centre}\ (-2,\ 3)\,,\ \text{radius}\sqrt{13}$.

Final answer

$\text{circle}\,;\ \text{centre}\ (-2,\ 3)\,,\ \text{radius}\sqrt{13}$

Worked Solutions **Circles**

Q019 MCQ

Determine what kind of figure $x^2 + y^2 - 6x + 2y + 6 = 0$ represents:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle ; centre $(3, -1)$, radius 2.

Correct option: A

Final answer

circle ; centre $(3, -1)$, radius 2

Worked Solutions **Circles**

Q020 *MCQ*

Determine what kind of figure $x^2 + y^2 - 4x - 6y + 4 = 0$ represents:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $\text{circle}\,;\ \text{centre}\ (2,\ 3)\ ,\ \text{radius}\,3$.

Correct option: C

Final answer

$\text{circle}\,;\ \text{centre}\ (2,\ 3)\ ,\ \text{radius}\,3$

Worked Solutions **Circles**

Q021 MCQ

Determine what kind of figure $x^2 + y^2 - 4x + 8y + 4 = 0$ represents:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $\text{circle; centre } (2, -4), \text{ radius } 4$.

Correct option: D

Final answer

$\text{circle; centre } (2, -4), \text{ radius } 4$

Worked Solutions **Circles**

Q022 *MCQ*

Determine what kind of figure $x^2 + y^2 - 4x = 0$ represents:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $\text{circle}; \text{centre } (2, 0), \text{ radius } 2$.

Correct option: C

Final answer

$\text{circle}; \text{centre } (2, 0), \text{ radius } 2$

Worked Solutions **Circles**

Q023 MCQ

Determine what kind of figure $x^2 + y^2 + 2x - 4y - 20 = 0$ represents:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $\text{circle}; \text{centre } (-1, 2), \text{ radius } 5$.

Correct option: C

Final answer

$\text{circle}; \text{centre } (-1, 2), \text{ radius } 5$

Worked Solutions **Circles**

Q024 MCQ

Determine what kind of figure $x^2 + y^2 + 4x - 6y = 0$ represents:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $\text{circle}\,;\ \text{centre}\ (-2,\ 3)\,,\ \text{radius}\sqrt{13}$.

Correct option: C

Final answer

$\text{circle}\,;\ \text{centre}\ (-2,\ 3)\,,\ \text{radius}\sqrt{13}$

Worked Solutions **Circles**

Q025 Challenge

A circle passes through A $(2,\ 5)$, B $(8,\ 1)$, and has its centre on $x + y = 8$. Find its standard equation, centre and radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $\text{centre } (5,\ 3)\,,\ \text{radius}^2 = 13; (x-5)^2 + (y-3)^2 = 13.$

Final answer

$\text{centre } (5,\ 3)\,,\ \text{radius}^2 = 13; (x-5)^2 + (y-3)^2 = 13$

Worked Solutions **Circles**

Q026 Challenge

A circle passes through A $(2,\ 5)$, B $(8,\ 1)$, and has its centre on $x + y = 8$. Find its standard equation, centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $\text{centre }(5,\ 3)\text{, radius}^2 = 13; (x-5)^2 + (y-3)^2 = 13$.

Correct option: B

Final answer

$\text{centre }(5,\ 3)\text{, radius}^2 = 13; (x-5)^2 + (y-3)^2 = 13$

Worked Solutions **Circles**

Q027 Written

Find the equation of the circle with centre (2,3) passing through (4,6).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x-2)^2 + (y-3)^2 = 13$.

Final answer

$(x-2)^2 + (y-3)^2 = 13$

Q028 Written

Find the equation of the circle with centre (-3,1) passing through (4,-2).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x+3)^2 + (y-1)^2 = 58$.

Final answer

$(x+3)^2 + (y-1)^2 = 58$

Worked Solutions **Circles**

Q029 Written

Find the equation of the circle with centre (1,-3) through (-3,0).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x-1)^2+(y+3)^2=25$.

Final answer

$(x-1)^2+(y+3)^2=25$

Q030 Written

Find the equation of the circle with centre (-4,1) through (3,-2).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x+4)^2+(y-1)^2=58$.

Final answer

$(x+4)^2+(y-1)^2=58$

Worked Solutions **Circles**

Q031 **Written**

Find the equation of the circle with centre (2,-1) through the origin.

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x-2)^2+(y+1)^2=5$.

Final answer

$(x-2)^2+(y+1)^2=5$

Q032 **Written**

Find the equation of the circle with centre the origin through (4,-3).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2+y^2=25$.

Final answer

$x^2+y^2=25$

Worked Solutions **Circles**

Q033 *MCQ*

Find the equation of the circle with centre (2,3) passing through (4,6):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-2)^2+(y-3)^2=13$.

Correct option: C

Final answer

$(x-2)^2+(y-3)^2=13$

Q034 *MCQ*

Find the equation of the circle with centre (-3,1) passing through (4,-2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+3)^2+(y-1)^2=58$.

Correct option: A

Final answer

$(x+3)^2+(y-1)^2=58$

Worked Solutions **Circles**

Q035 MCQ

Find the equation of the circle with centre (1,-3) through (-3,0):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-1)^2+(y+3)^2=25$.

Correct option: D

Final answer

$(x-1)^2+(y+3)^2=25$

Q036 MCQ

Find the equation of the circle with centre (-4,1) through (3,-2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+4)^2+(y-1)^2=58$.

Correct option: A

Final answer

$(x+4)^2+(y-1)^2=58$

Worked Solutions **Circles**

Q037 MCQ

Find the equation of the circle with centre (2,-1) through the origin:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-2)^2+(y+1)^2=5$.

Correct option: B

Final answer

$(x-2)^2+(y+1)^2=5$

Q038 MCQ

Find the equation of the circle with centre the origin through (4,-3):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2+y^2=25$.

Correct option: A

Final answer

$x^2+y^2=25$

Q039 Written

Find the equation of the circle with diameter endpoints A $(1,\ 3)$, B $(3,\ -5)$.

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x-2)^2+(y+1)^2=17$.

Final answer

$(x-2)^2+(y+1)^2=17$

Q040 Written

Find the equation of the circle with diameter endpoints (3,4), (5,-2).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x-4)^2+(y-1)^2=10$.

Final answer

$(x-4)^2+(y-1)^2=10$

Worked Solutions **Circles**

Q041 **Written**

Find the equation of the circle with diameter endpoints (4,7), (-2,-1).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x-1)^2+(y-3)^2=25$.

Final answer

$(x-1)^2+(y-3)^2=25$

Q042 **Written**

Find the equation of the circle with diameter endpoints (4,-2), (-6,-2).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x+1)^2+(y+2)^2=25$.

Final answer

$(x+1)^2+(y+2)^2=25$

Worked Solutions **Circles**

Q043 Written

Find the equation of the circle with diameter endpoints (-3,0), (-1,2).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x+2)^2+(y-1)^2=2$.

Final answer

$(x+2)^2+(y-1)^2=2$

Q044 Written

Find the equation of the circle with diameter endpoints (-3,6), (3,-2).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2+(y-2)^2=25$.

Final answer

$x^2+(y-2)^2=25$

Worked Solutions **Circles**

Q045 *MCQ*

Find the equation of the circle with diameter endpoints A $(1, 3)$, B $(3, -5)$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-2)^2 + (y+1)^2 = 17$.

Correct option: A

Final answer

$(x-2)^2 + (y+1)^2 = 17$

Q046 *MCQ*

Find the equation of the circle with diameter endpoints (3,4), (5,-2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-4)^2 + (y-1)^2 = 10$.

Correct option: C

Final answer

$(x-4)^2 + (y-1)^2 = 10$

Worked Solutions **Circles**

Q047 MCQ

Find the equation of the circle with diameter endpoints (4,7), (-2,-1):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-1)^2+(y-3)^2=25$.

Correct option: C

Final answer

$(x-1)^2+(y-3)^2=25$

Q048 MCQ

Find the equation of the circle with diameter endpoints (4,-2), (-6,-2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+1)^2+(y+2)^2=25$.

Correct option: A

Final answer

$(x+1)^2+(y+2)^2=25$

Worked Solutions **Circles**

Q049 MCQ

Find the equation of the circle with diameter endpoints (-3,0), (-1,2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+2)^2 + (y-1)^2 = 2$.

Correct option: B

Final answer

$(x+2)^2 + (y-1)^2 = 2$

Q050 MCQ

Find the equation of the circle with diameter endpoints (-3,6), (3,-2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + (y-2)^2 = 25$.

Correct option: C

Final answer

$x^2 + (y-2)^2 = 25$

Worked Solutions **Circles**

Q051 Written

Find the equation of the circle passing through (1,1), (2,-1), and (3,2).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 5x - y + 4 = 0$.

Final answer

$x^2 + y^2 - 5x - y + 4 = 0$

Q052 Written

Find the equation of the circle passing through (0,1), (-1,2), and (3,1).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 3x - 7y + 6 = 0$.

Final answer

$x^2 + y^2 - 3x - 7y + 6 = 0$

Worked Solutions **Circles**

Q053 Written

Find the equation of the circle with through (0,0), (-1,1), (2,4).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 2x - 4y = 0$.

Final answer

$x^2 + y^2 - 2x - 4y = 0$

Q054 Written

Find the equation of the circle with through (-4,6), (0,8), (3,-1).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 6y - 16 = 0$.

Final answer

$x^2 + y^2 - 6y - 16 = 0$

Worked Solutions **Circles**

Q055 Written

Find the equation of the circle with through (1,1), (5,-1), (-3,-7).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 2x + 8y - 8 = 0$.

Final answer

$x^2 + y^2 - 2x + 8y - 8 = 0$

Q056 MCQ

Find the equation of the circle passing through (1,1), (2,-1), and (3,2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 5x - y + 4 = 0$.

Correct option: A

Final answer

$x^2 + y^2 - 5x - y + 4 = 0$

Worked Solutions **Circles**

Q057 *MCQ*

Find the equation of the circle passing through (0,1), (-1,2), and (3,1):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 3x - 7y + 6 = 0$.

Correct option: B

Final answer

$x^2 + y^2 - 3x - 7y + 6 = 0$

Q058 *MCQ*

Find the equation of the circle with through (0,0), (-1,1), (2,4):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 2x - 4y = 0$.

Correct option: D

Final answer

$x^2 + y^2 - 2x - 4y = 0$

Worked Solutions **Circles**

Q059 MCQ

Find the equation of the circle with through (-4,6), (0,8), (3,-1):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 6y - 16 = 0$.

Correct option: D

Final answer

$x^2 + y^2 - 6y - 16 = 0$

Q060 MCQ

Find the equation of the circle with through (1,1), (5,-1), (-3,-7):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 2x + 8y - 8 = 0$.

Correct option: D

Final answer

$x^2 + y^2 - 2x + 8y - 8 = 0$

Worked Solutions **Circles**

Q061 **Challenge**

Find the circle through (0,2), (2,0), and (4,6), giving its centre and radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $x^2 + y^2 - 6x - 6y + 8 = 0$; centre $(3,\ 3)$, radius $\sqrt{10}$.

Final answer

$x^2 + y^2 - 6x - 6y + 8 = 0$; centre $(3,\ 3)$, radius $\sqrt{10}$

Worked Solutions **Circles**

Q062 Challenge

Find the circle through (0,2), (2,0), and (4,6), giving its centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 6x - 6y + 8 = 0$; centre $(3,\ 3)$, radius $\sqrt{10}$.

Correct option: A

Final answer

$x^2 + y^2 - 6x - 6y + 8 = 0$; centre $(3,\ 3)$, radius $\sqrt{10}$

Worked Solutions **Circles**

Q063 Written

Find the circle with centre (4,-2) tangent to the x-axis.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x-4)^2+(y+2)^2=4$.

Final answer

$(x-4)^2+(y+2)^2=4$

Q064 Written

Find the circle with centre (-2,3) tangent to the y-axis.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x+2)^2+(y-3)^2=4$.

Final answer

$(x+2)^2+(y-3)^2=4$

Worked Solutions **Circles**

Q065 **Written**

Find the circle with centre (-1,4) tangent to x-axis.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x+1)^2 + (y-4)^2 = 16$.

Final answer

$(x+1)^2 + (y-4)^2 = 16$

Q066 **Written**

Find the circle with centre (3,-4) tangent to x-axis.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x-3)^2 + (y+4)^2 = 16$.

Final answer

$(x-3)^2 + (y+4)^2 = 16$

Worked Solutions **Circles**

Q067 **Written**

Find the circle with centre (5,-4) tangent to x-axis.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x-5)^2+(y+4)^2=16$.

Final answer

$(x-5)^2+(y+4)^2=16$

Q068 **Written**

Find the circle with centre (-1,6) tangent to y-axis.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x+1)^2+(y-6)^2=1$.

Final answer

$(x+1)^2+(y-6)^2=1$

Worked Solutions **Circles**

Q069 Written

Find the circle with centre (2,-5) tangent to y-axis.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x-2)^2 + (y+5)^2 = 4$.

Final answer

$(x-2)^2 + (y+5)^2 = 4$

Q070 Written

Find the circle with centre (-3,-4) tangent to y-axis.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x+3)^2 + (y+4)^2 = 9$.

Final answer

$(x+3)^2 + (y+4)^2 = 9$

Worked Solutions **Circles**

Q071 *MCQ*

Find the circle with centre (4,-2) tangent to the x-axis:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-4)^2+(y+2)^2=4$.

Correct option: B

Final answer

$(x-4)^2+(y+2)^2=4$

Worked Solutions **Circles**

Q072 MCQ

Find the circle with centre (-2,3) tangent to the y-axis:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+2)^2 + (y-3)^2 = 4$.

Correct option: B

Final answer

$(x+2)^2 + (y-3)^2 = 4$

Worked Solutions **Circles**

Q073 *MCQ*

Find the circle with centre (-1,4) tangent to x-axis:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+1)^2+(y-4)^2=16$.

Correct option: B

Final answer

$(x+1)^2+(y-4)^2=16$

Q074 MCQ

Find the circle with centre (3,-4) tangent to x-axis:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-3)^2+(y+4)^2=16$.

Correct option: D

Final answer

$(x-3)^2+(y+4)^2=16$

Worked Solutions **Circles**

Q075 MCQ

Find the circle with centre (5,-4) tangent to x-axis:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-5)^2+(y+4)^2=16$.

Correct option: D

Final answer

$(x-5)^2+(y+4)^2=16$

Worked Solutions **Circles**

Q076 MCQ

Find the circle with centre (-1,6) tangent to y-axis:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+1)^2+(y-6)^2=1$.

Correct option: D

Final answer

$(x+1)^2+(y-6)^2=1$

Worked Solutions **Circles**

Q077 MCQ

Find the circle with centre (2,-5) tangent to y-axis:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-2)^2+(y+5)^2=4$.

Correct option: B

Final answer

$(x-2)^2+(y+5)^2=4$

Worked Solutions **Circles**

Q078 MCQ

Find the circle with centre (-3,-4) tangent to y-axis:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+3)^2+(y+4)^2=9$.

Correct option: C

Final answer

$(x+3)^2+(y+4)^2=9$

Worked Solutions **Circles**

Q079 Written

Find the circles through (-2,1) tangent to both the x-axis and y-axis.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x+1)^2+(y-1)^2=1$ or $(x+5)^2+(y-5)^2=25$.

Final answer

$(x+1)^2+(y-1)^2=1$ or $(x+5)^2+(y-5)^2=25$

Q080 Written

Find the circles through (1,2) tangent to both axes.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x-1)^2+(y-1)^2=1$ or $(x-5)^2+(y-5)^2=25$.

Final answer

$(x-1)^2+(y-1)^2=1$ or $(x-5)^2+(y-5)^2=25$

Worked Solutions **Circles**

Q081 Written

Find the circle with through (-1,-2) tangent to both axes.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x+1)^2+(y+1)^2=1$ or $(x+5)^2+(y+5)^2=25$.

Final answer

$(x+1)^2+(y+1)^2=1$ or $(x+5)^2+(y+5)^2=25$

Q082 Written

Find the circle with through (-2,-1) tangent to both axes.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x+1)^2+(y+1)^2=1$ or $(x+5)^2+(y+5)^2=25$.

Final answer

$(x+1)^2+(y+1)^2=1$ or $(x+5)^2+(y+5)^2=25$

Worked Solutions **Circles**

Q083 Written

Find the circle with through (8,-1) tangent to both axes.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x-5)^2+(y+5)^2=25$ or $(x-13)^2+(y+13)^2=169$.

Final answer

$(x-5)^2+(y+5)^2=25$ or $(x-13)^2+(y+13)^2=169$

Worked Solutions **Circles**

Q084 MCQ

Find the circles through (-2,1) tangent to both the x-axis and y-axis:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+1)^2+(y-1)^2=1$ or $(x+5)^2+(y-5)^2=25$.

Correct option: C

Final answer

$(x+1)^2+(y-1)^2=1$ or $(x+5)^2+(y-5)^2=25$

Worked Solutions **Circles**

Q085 MCQ

Find the circles through (1,2) tangent to both axes:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-1)^2+(y-1)^2=1$ or $(x-5)^2+(y-5)^2=25$.

Correct option: D

Final answer

$(x-1)^2+(y-1)^2=1$ or $(x-5)^2+(y-5)^2=25$

Worked Solutions **Circles**

Q086 MCQ

Find the circle with through (-1,-2) tangent to both axes:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+1)^2+(y+1)^2=1$ or $(x+5)^2+(y+5)^2=25$.

Correct option: C

Final answer

$(x+1)^2+(y+1)^2=1$ or $(x+5)^2+(y+5)^2=25$

Worked Solutions **Circles**

Q087 MCQ

Find the circle with through (-2,-1) tangent to both axes:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$.

Correct option: C

Final answer

$(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$

Worked Solutions **Circles**

Q088 MCQ

Find the circle with through (8,-1) tangent to both axes:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-5)^2+(y+5)^2=25$ or $(x-13)^2+(y+13)^2=169$.

Correct option: B

Final answer

$(x-5)^2+(y+5)^2=25$ or $(x-13)^2+(y+13)^2=169$

Worked Solutions **Circles**

Q089 Written

Find the circle whose centre is on $y = -x$ and which passes through (5,2) and (-2,3).

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x-1)^2 + (y+1)^2 = 25$.

Final answer

$(x-1)^2 + (y+1)^2 = 25$

Q090 Written

Find the circle whose centre is on $y = x + 5$ and which passes through the origin and (1,2).

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x+5/2)^2 + (y-5/2)^2 = 25/2$.

Final answer

$(x+5/2)^2 + (y-5/2)^2 = 25/2$

Worked Solutions **Circles**

Q091 Written

Find the circle with centre on $y = x + 1$ through (3,4) and (1,4).

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x-2)^2 + (y-3)^2 = 2$.

Final answer

$(x-2)^2 + (y-3)^2 = 2$

Q092 Written

Find the circle with centre on $y = 2x - 6$ through (3,7) and (-1,3).

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x-4)^2 + (y-2)^2 = 26$.

Final answer

$(x-4)^2 + (y-2)^2 = 26$

Worked Solutions **Circles**

Q093 MCQ

Find the circle whose centre is on $y = -x$ and which passes through (5,2) and (-2,3):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-1)^2 + (y+1)^2 = 25$.

Correct option: D

Final answer

$(x-1)^2 + (y+1)^2 = 25$

Worked Solutions **Circles**

Q094 MCQ

Find the circle whose centre is on $y = x + 5$ and which passes through the origin and (1,2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 5/2)^2 + (y - 5/2)^2 = 25/2$.

Correct option: C

Final answer

$(x + 5/2)^2 + (y - 5/2)^2 = 25/2$

Worked Solutions **Circles**

Q095 MCQ

Find the circle with centre on $y = x + 1$ through (3,4) and (1,4):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-2)^2 + (y-3)^2 = 2$.

Correct option: C

Final answer

$(x-2)^2 + (y-3)^2 = 2$

Worked Solutions **Circles**

Q096 MCQ

Find the circle with centre on $y = 2x - 6$ through (3,7) and (-1,3):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-4)^2 + (y-2)^2 = 26$.

Correct option: D

Final answer

$(x-4)^2 + (y-2)^2 = 26$

Worked Solutions **Circles**

Q097 Challenge

A circle centre is constrained to the x-axis and passes through P1(-2,2), P2(1,1). Find its standard equation, centre and radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $(x+1)^2+y^2=5$; centre $(-1,\ 0)$, radius $\sqrt{5}$.

Final answer

$(x+1)^2+y^2=5$; centre $(-1,\ 0)$, radius $\sqrt{5}$

Worked Solutions **Circles**

Q098 Challenge

A circle centre is constrained to the x-axis and passes through P1(-2,2), P2(1,1). Find its standard equation, centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5}$.

Correct option: C

Final answer

$(x+1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5}$

Worked Solutions **Circles**

Q099 **Written**

Find the range of k for $x^2 + y^2 - 2\,\mathbf{kx} + 2\,\mathbf{ky} + 4k + 6 = 0$ to represent a circle.

Solution

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < -1$ or $k > 3$.

Final answer

$k < -1$ or $k > 3$

Q100 **Written**

Find the range of k for $x^2 + y^2 + 2\,\mathbf{kx} - 4\,\mathbf{ky} + 4k^2 + 6 = 0$ to represent a circle.

Solution

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < -\sqrt{6}$ or $k > \sqrt{6}$.

Final answer

$k < -\sqrt{6}$ or $k > \sqrt{6}$

Worked Solutions **Circles**

Q101 Written

Find the range of k for $x^2 + y^2 + 2x - y + k = 0$ to represent a circle.

Solution

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < 5/4$.

Final answer

$k < 5/4$

Q102 Written

Find the range of k for $x^2 + y^2 + 6\,\mathbf{kx} - 2\,\mathbf{ky} + 28k + 6 = 0$ to represent a circle.

Solution

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < -1/5 \,\text{or}\, k > 3$.

Final answer

$k < -1/5 \,\text{or}\, k > 3$

Worked Solutions **Circles**

Q103 Written

Find the range of k for $x^2 + y^2 + 2\mathbf{kx} - 2(k+3)y + 4k^2 + 1 = 0$ to represent a circle.

Solution

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $-1 < k < 4$.

Final answer

$-1 < k < 4$

Worked Solutions **Circles**

Q104 MCQ

Find the range of k for $x^2 + y^2 - 2\,\mathbf{kx} + 2\,\mathbf{ky} + 4k + 6 = 0$ to represent a circle:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < -1$ or $k > 3$.

Correct option: C

Final answer

$k < -1$ or $k > 3$

Worked Solutions **Circles**

Q105 MCQ

Find the range of k for $x^2 + y^2 + 2\mathbf{kx} - 4\mathbf{ky} + 4k^2 + 6 = 0$ to represent a circle:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < -\sqrt{6}$ or $k > \sqrt{6}$.

Correct option: B

Final answer

$k < -\sqrt{6}$ or $k > \sqrt{6}$

Worked Solutions **Circles**

Q106 MCQ

Find the range of k for $x^2 + y^2 + 2x - y + k = 0$ to represent a circle:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < 5/4$.

Correct option: C

Final answer

$k < 5/4$

Worked Solutions **Circles**

Q107 MCQ

Find the range of k for $x^2 + y^2 + 6\,\mathbf{kx} - 2\,\mathbf{ky} + 28k + 6 = 0$ to represent a circle:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < -1/5$ or $k > 3$.

Correct option: B

Final answer

$k < -1/5$ or $k > 3$

Worked Solutions **Circles**

Q108 MCQ

Find the range of k for $x^2 + y^2 + 2\,\mathbf{kx} - 2(k+3)y + 4k^2 + 1 = 0$ to represent a circle:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-1 < k < 4$.

Correct option: D

Final answer

$-1 < k < 4$

Worked Solutions **Circles**

Q109 **Challenge**

An acoustic-safety circle is $x^2 + y^2 + 6x - 2\mathbf{ky} + 2k^2 - 4k + 9 = 0$. Find the exact k-range giving a positive radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $0 < k < 4$.

Final answer

$0 < k < 4$

Worked Solutions **Circles**

Q110 Challenge

An acoustic-safety circle is $x^2 + y^2 + 6x - 2\mathbf{ky} + 2k^2 - 4k + 9 = 0$. Find the exact k-range giving a positive radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $0 < k < 4$.

Correct option: B

Final answer

$0 < k < 4$

Worked Solutions **Circles**

Q111 Written

Find the coordinates of the intersections of $(x + 2)^2 + y^2 = 10$ and $y = -2x + 1$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(1, -1)$ and $(-1, 3)$.

Final answer

$(1, -1)$ and $(-1, 3)$

Q112 Written

Find the coordinates of the intersections of $x^2 + y^2 = 4$ and $x + y = 2$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(0, 2)$ and $(2, 0)$.

Final answer

$(0, 2)$ and $(2, 0)$

Q113 Written

Find the coordinates of the intersections of $x^2 + y^2 = 10$ with $y = 3x - 10$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(3, -1)$.

Final answer

$(3, -1)$

Q114 Written

Find the coordinates of the intersections of $x^2 + y^2 = 5$ with $2x - y + 5 = 0$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(-2, 1)$.

Final answer

$(-2, 1)$

Worked Solutions **Circles**

Q115 **Written**

Find the coordinates of the intersections of $(x-1)^2+y^2=10$ with $x+3y-1=0$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(-2, 1)$ and $(4, -1)$.

Final answer

$(-2, 1)$ and $(4, -1)$

Worked Solutions **Circles**

Q116 **MCQ**

Find the coordinates of the intersections of $(x+2)^2 + y^2 = 10$ and $y = -2x + 1$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(1, -1)$ and $(-1, 3)$.

Correct option: C

Final answer

$(1, -1)$ and $(-1, 3)$

Worked Solutions **Circles**

Q117 *MCQ*

Find the coordinates of the intersections of $x^2 + y^2 = 4$ and $x + y = 2$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is (0, 2) and (2, 0).

Correct option: D

Final answer

(0, 2) and (2, 0)

Worked Solutions **Circles**

Q118 **MCQ**

Find the coordinates of the intersections of $x^2 + y^2 = 10$ with $y = 3x - 10$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(3, -1)$.

Correct option: C

Final answer

$(3, -1)$

Worked Solutions **Circles**

Q119 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 5$ with $2x - y + 5 = 0$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-2, 1)$.

Correct option: D

Final answer

$(-2, 1)$

Worked Solutions **Circles**

Q120 MCQ

Find the coordinates of the intersections of $(x-1)^2 + y^2 = 10$ with $x + 3y - 1 = 0$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-2, 1)$ and $(4, -1)$.

Correct option: B

Final answer

$(-2, 1)$ and $(4, -1)$

Worked Solutions **Circles**

Q121 **Written**

Find the number of intersections of $x^2 + y^2 = 2$ and $x - y + 3 = 0$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $0\,\text{points}$.

Final answer

$0\,\text{points}$

Q122 **Written**

Find the number of intersections of $x^2 + y^2 = 10$ and $y = 2x - 5$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $2\,\text{points}\ :\ (1, -3)$ and $(3, 1)$.

Final answer

$2\,\text{points}\ :\ (1, -3)$ and $(3, 1)$

Worked Solutions **Circles**

Q123 **Written**

Find the number of intersections of $x^2 + y^2 = 5$ with $y = x + 2$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $2\,\text{points}\ :\ \left(-1+\sqrt{6}/2,\ 1+\sqrt{6}/2\right)$ and $\left(-1-\sqrt{6}/2,\ 1-\sqrt{6}/2\right)$.

Final answer

$2\,\text{points}\ :\ \left(-1+\sqrt{6}/2,\ 1+\sqrt{6}/2\right)$ and $\left(-1-\sqrt{6}/2,\ 1-\sqrt{6}/2\right)$

Q124 **Written**

Find the number of intersections of $x^2 + y^2 = 2$ with $2x + 3y - 6 = 0$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $0\,\text{points}$.

Final answer

$0\,\text{points}$

Worked Solutions **Circles**

Q125 **Written**

Find the number of intersections of $(x+1)^2+(y-2)^2=5$ with $x+2y+2=0$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $1\,\text{point}\ :\ (-2,\,0)$.

Final answer

$1\,\text{point}\ :\ (-2,\,0)$

Worked Solutions **Circles**

Q126 MCQ

Find the number of intersections of $x^2 + y^2 = 2$ and $x - y + 3 = 0$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $0\,\text{points}$.

Correct option: D

Final answer

$0\,\text{points}$

Worked Solutions **Circles**

Q127 MCQ

Find the number of intersections of $x^2 + y^2 = 10$ and $y = 2x - 5$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2\,\text{points}\ :\ (1, -3)$ and $(3, 1)$.

Correct option: A

Final answer

$2\,\text{points}\ :\ (1, -3)$ and $(3, 1)$

Worked Solutions **Circles**

Q128 MCQ

Find the number of intersections of $x^2 + y^2 = 5$ with $y = x + 2$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2\,\text{points}\ :\ \left(-1 + \sqrt{6}/2,\ 1 + \sqrt{6}/2\right)$ and $\left(-1 - \sqrt{6}/2,\ 1 - \sqrt{6}/2\right)$.

Correct option: C

Final answer

$2\,\text{points}\ :\ \left(-1 + \sqrt{6}/2,\ 1 + \sqrt{6}/2\right)$ and $\left(-1 - \sqrt{6}/2,\ 1 - \sqrt{6}/2\right)$

Worked Solutions **Circles**

Q129 MCQ

Find the number of intersections of $x^2 + y^2 = 2$ with $2x + 3y - 6 = 0$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $0\,\text{points}$.

Correct option: C

Final answer

$0\,\text{points}$

Worked Solutions **Circles**

Q130 MCQ

Find the number of intersections of $(x+1)^2+(y-2)^2=5$ with $x+2y+2=0$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $1\,\text{point} \;:\; (-2,\,0)$.

Correct option: D

Final answer

$1\,\text{point} \;:\; (-2,\,0)$

Worked Solutions **Circles**

Q131 Written

If $x^2 + y^2 = 5$ and $y = 2x - k$ intersect at two distinct points, find the range of k.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $-5 < k < 5$.

Final answer

$-5 < k < 5$

Q132 Written

If $x^2 + y^2 = r^2$ and $2x + y - 5 = 0$ are tangent, find r.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $r = \sqrt{5}$.

Final answer

$r = \sqrt{5}$

Worked Solutions **Circles**

Q133 Written

If $x^2 + y^2 = 4$ and $y = \text{mx} + 4$ are tangent, find m.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $m = \pm\sqrt{3}$.

Final answer

$m = \pm\sqrt{3}$

Q134 Written

If $x^2 + y^2 = r^2$ and $x - 3y - 10 = 0$ have no intersections, find the positive-radius range.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $0 < r < \sqrt{10}$.

Final answer

$0 < r < \sqrt{10}$

Worked Solutions **Circles**

Q135 Written

If $x^2 + y^2 = 8$ and $y = x + m$ are tangent, find m.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $m = \pm 4$.

Final answer

$m = \pm 4$

Q136 Written

If $x^2 + y^2 = 10$ and $y = 3x + n$ have no intersections, find n.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $n < -10 \text{ or } n > 10$.

Final answer

$n < -10 \text{ or } n > 10$

Worked Solutions **Circles**

Q137 Written

If $x^2 + y^2 = 5$ and $y = 2x + m$ have points of intersection, find the range of m.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $-5 <= m <= 5$.

Final answer

$-5 <= m <= 5$

Q138 Written

If $x^2 + y^2 = r^2$ and $y = x + 2$ are tangent, find r.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $r = \sqrt{2}$.

Final answer

$r = \sqrt{2}$

Worked Solutions **Circles**

Q139 **Written**

If $x^2 + y^2 = r^2$ and $4x - y + 17 = 0$ intersect at two distinct points, find the positive-radius range.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $r > \sqrt{17}$.

Final answer

$r > \sqrt{17}$

Worked Solutions **Circles**

Q140 MCQ

If $x^2 + y^2 = 5$ and $y = 2x - k$ intersect at two distinct points, find the range of k:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-5 < k < 5$.

Correct option: D

Final answer

$-5 < k < 5$

Worked Solutions **Circles**

Q141 MCQ

If $x^2 + y^2 = r^2$ and $2x + y - 5 = 0$ are tangent, find r:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $r = \sqrt{5}$.

Correct option: B

Final answer

$r = \sqrt{5}$

Worked Solutions **Circles**

Q142 MCQ

If $x^2 + y^2 = 4$ and $y = \mathbf{mx} + 4$ are tangent, find m:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $m = \pm\sqrt{3}$.

Correct option: B

Final answer

$m = \pm\sqrt{3}$

Worked Solutions **Circles**

Q143 MCQ

If $x^2 + y^2 = r^2$ and $x - 3y - 10 = 0$ have no intersections, find the positive-radius range:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $0 < r < \sqrt{10}$.

Correct option: B

Final answer

$0 < r < \sqrt{10}$

Worked Solutions **Circles**

Q144 MCQ

If $x^2 + y^2 = 8$ and $y = x + m$ are tangent, find m:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $m = \pm 4$.

Correct option: C

Final answer

$m = \pm 4$

Worked Solutions **Circles**

Q145 MCQ

If $x^2 + y^2 = 10$ and $y = 3x + n$ have no intersections, find n:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $n < -10$ or $n > 10$.

Correct option: A

Final answer

$n < -10$ or $n > 10$

Worked Solutions **Circles**

Q146 MCQ

If $x^2 + y^2 = 5$ and $y = 2x + m$ have points of intersection, find the range of m:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-5 <= m <= 5$.

Correct option: C

Final answer

$-5 <= m <= 5$

Worked Solutions **Circles**

Q147 MCQ

If $x^2 + y^2 = r^2$ and $y = x + 2$ are tangent, find r:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $r = \sqrt{2}$.

Correct option: B

Final answer

$r = \sqrt{2}$

Worked Solutions **Circles**

Q148 MCQ

If $x^2 + y^2 = r^2$ and $4x - y + 17 = 0$ intersect at two distinct points, find the positive-radius range:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $r > \sqrt{17}$.

Correct option: D

Final answer

$r > \sqrt{17}$

Worked Solutions **Circles**

Q149 **Challenge**

A radar circle $x^2 + y^2 = r^2$ must have no intersection with $3x + 4y - 10 = 0$. Find the positive range of r.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $0 < r < 2$.

Final answer

$0 < r < 2$

Worked Solutions **Circles**

Q150 **Challenge**

A radar circle $x^2 + y^2 = r^2$ must have no intersection with $3x + 4y - 10 = 0$. Find the positive range of r:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $0 < r < 2$.

Correct option: D

Final answer

$0 < r < 2$

Worked Solutions **Circles**

Q151 Written

Find the chord length cut off by $x^2 + y^2 = 4$ on $x + y - 1 = 0$.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $\sqrt{14}$.

Final answer

$\sqrt{14}$

Q152 Written

Find the chord length cut off by $(x - 2)^2 + (y + 1)^2 = 9$ on $x + 2y - 5 = 0$.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: 4.

Final answer

4

Worked Solutions **Circles**

Q153 **Written**

Find the chord length cut off by $x^2 + y^2 = 10$ on $2x - y = 5$.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $2\sqrt{5}$.

Final answer

$2\sqrt{5}$

Q154 **Written**

Find the chord length cut off by $(x - 2)^2 + (y - 1)^2 = 5$ on $x + y = 4$.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $3\sqrt{2}$.

Final answer

$3\sqrt{2}$

Worked Solutions **Circles**

Q155 Written

Find the chord length cut off by $x^2 + y^2 = 3$ on $x - y = 2$.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: 2.

Final answer

2

Q156 MCQ

Find the chord length cut off by $x^2 + y^2 = 4$ on $x + y - 1 = 0$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $\sqrt{14}$.

Correct option: A

Final answer

$\sqrt{14}$

Worked Solutions **Circles**

Q157 MCQ

Find the chord length cut off by $(x-2)^2+(y+1)^2=9$ on $x+2y-5=0$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is 4.

Correct option: C

Final answer

4

Q158 MCQ

Find the chord length cut off by $x^2+y^2=10$ on $2x-y=5$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2\sqrt{5}$.

Correct option: D

Final answer

$2\sqrt{5}$

Worked Solutions **Circles**

Q159 *MCQ*

Find the chord length cut off by $(x-2)^2+(y-1)^2=5$ on $x+y=4$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $3\sqrt{2}$.

Correct option: B

Final answer

$3\sqrt{2}$

Q160 *MCQ*

Find the chord length cut off by $x^2+y^2=3$ on $x-y=2$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is 2.

Correct option: D

Final answer

2

Worked Solutions **Circles**

Q161 Written

For $x^2 + y^2 = 3$ and $x - y + k = 0$, find k when the chord length is 2.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = \pm 2$.

Final answer

$k = \pm 2$

Q162 Written

For $x^2 + y^2 = 16$ and $y = x + k$, find k when the chord length is sqrt(2).

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = \pm\sqrt{31}$.

Final answer

$k = \pm\sqrt{31}$

Worked Solutions **Circles**

Q163 Written

For $x^2 + y^2 = 10$ and $y = x + k$, find k when the chord length is 4sqrt(2).

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = \pm 2$.

Final answer

$k = \pm 2$

Q164 Written

For $(x-1)^2 + y^2 = 5$ and $y = x + k$, find k when the chord length is 3sqrt(2).

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = 0 \text{ or } k = -2$.

Final answer

$k = 0 \text{ or } k = -2$

Worked Solutions **Circles**

Q165 MCQ

For $x^2 + y^2 = 3$ and $x - y + k = 0$, find k when the chord length is 2:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = \pm 2$.

Correct option: C

Final answer

$k = \pm 2$

Q166 MCQ

For $x^2 + y^2 = 16$ and $y = x + k$, find k when the chord length is sqrt(2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = \pm\sqrt{31}$.

Correct option: D

Final answer

$k = \pm\sqrt{31}$

Worked Solutions **Circles**

Q167 MCQ

For $x^2 + y^2 = 10$ and $y = x + k$, find k when the chord length is 4sqrt(2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = \pm 2$.

Correct option: A

Final answer

$k = \pm 2$

Worked Solutions **Circles**

Q168 MCQ

For $(x-1)^2 + y^2 = 5$ and $y = x + k$, find k when the chord length is 3sqrt(2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = 0$ or $k = -2$.

Correct option: B

Final answer

$k = 0$ or $k = -2$

Worked Solutions **Circles**

Q169 **Challenge**

For $x^2 + y^2 + 4x - 2y - 7 = 0$ and $y = x + k$, find k when the chord length is 4.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $k = -1$ or $k = 7$.

Final answer

$k = -1$ or $k = 7$

Worked Solutions **Circles**

Q170 Challenge

For $x^2 + y^2 + 4x - 2y - 7 = 0$ and $y = x + k$, find k when the chord length is 4:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = -1$ or $k = 7$.

Correct option: A

Final answer

$k = -1$ or $k = 7$

Worked Solutions **Circles**

Q171 **Written**

Find the tangent to $x^2 + y^2 = 25$ at (4,3).

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $4x + 3y = 25$.

Final answer

$4x + 3y = 25$

Q172 **Written**

Find the tangent to $x^2 + y^2 = 34$ at (5,3).

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $5x + 3y = 34$.

Final answer

$5x + 3y = 34$

Worked Solutions **Circles**

Q173 Written

Find the tangents $x^2 + y^2 = 10$ at (3,-1).

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $3x - y = 10$.

Final answer

$3x - y = 10$

Q174 Written

Find the tangents $x^2 + y^2 = 29$ at (-5,-2).

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-5x - 2y = 29$.

Final answer

$-5x - 2y = 29$

Worked Solutions **Circles**

Q175 Written

Find the tangents $x^2 + y^2 = 5$ at (-2,1).

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-2x + y = 5$.

Final answer

$-2x + y = 5$

Worked Solutions **Circles**

Q176 MCQ

Find the tangent to $x^2 + y^2 = 25$ at (4,3):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $4x + 3y = 25$.

Correct option: A

Final answer

$4x + 3y = 25$

Worked Solutions **Circles**

Q177 MCQ

Find the tangent to $x^2 + y^2 = 34$ at (5,3):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $5x + 3y = 34$.

Correct option: C

Final answer

$5x + 3y = 34$

Worked Solutions **Circles**

Q178 MCQ

Find the tangents $x^2 + y^2 = 10$ at (3,-1):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $3x - y = 10$.

Correct option: A

Final answer

$3x - y = 10$

Worked Solutions **Circles**

Q179 MCQ

Find the tangents $x^2 + y^2 = 29$ at (-5,-2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-5x - 2y = 29$.

Correct option: C

Final answer

$-5x - 2y = 29$

Worked Solutions **Circles**

Q180 MCQ

Find the tangents $x^2 + y^2 = 5$ at (-2,1):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-2x + y = 5$.

Correct option: A

Final answer

$-2x + y = 5$

Worked Solutions **Circles**

Q181 Written

Find the tangents from (1,3) to $x^2 + y^2 = 5$.

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x-h)(x_1-h)+(y-k)(y_1-k)=r^2$ and rearrange.
3. Tangent: $2x+y=5$ and $-x+2y=5$.

Final answer

$2x+y=5$ and $-x+2y=5$

Q182 Written

Find the tangents from (-2,4) to $x^2 + y^2 = 10$.

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x-h)(x_1-h)+(y-k)(y_1-k)=r^2$ and rearrange.
3. Tangent: $-3x+y=10$ and $x+3y=10$.

Final answer

$-3x+y=10$ and $x+3y=10$

Worked Solutions **Circles**

Q183 **Written**

Find the tangents from (-1,3) to $x^2 + y^2 = 5$.

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x-h)(x_1-h)+(y-k)(y_1-k)=r^2$ and rearrange.
3. Tangent: $-2x + y = 5$ and $x + 2y = 5$.

Final answer

$-2x + y = 5$ and $x + 2y = 5$

Q184 **Written**

Find the tangents from (2,1) to $x^2 + y^2 = 1$.

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x-h)(x_1-h)+(y-k)(y_1-k)=r^2$ and rearrange.
3. Tangent: $y = 1$ and $4x - 3y = 5$.

Final answer

$y = 1$ and $4x - 3y = 5$

Worked Solutions **Circles**

Q185 **Written**

Find the tangents from (2,6) to $x^2 + y^2 = 20$.

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x-h)(x_1-h)+(y-k)(y_1-k)=r^2$ and rearrange.
3. Tangent: $-x+2y=10$ and $2x+y=10$.

Final answer

$-x+2y=10$ and $2x+y=10$

Worked Solutions **Circles**

Q186 **MCQ**

Find the tangents from (1,3) to $x^2 + y^2 = 5$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2x + y = 5$ and $-x + 2y = 5$.

Correct option: D

Final answer

$2x + y = 5$ and $-x + 2y = 5$

Worked Solutions **Circles**

Q187 MCQ

Find the tangents from (-2,4) to $x^2 + y^2 = 10$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-3x + y = 10$ and $x + 3y = 10$.

Correct option: A

Final answer

$-3x + y = 10$ and $x + 3y = 10$

Worked Solutions **Circles**

Q188 **MCQ**

Find the tangents from (-1,3) to $x^2 + y^2 = 5$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-2x + y = 5$ and $x + 2y = 5$.

Correct option: D

Final answer

$-2x + y = 5$ and $x + 2y = 5$

Worked Solutions **Circles**

Q189 MCQ

Find the tangents from (2,1) to $x^2 + y^2 = 1$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $y = 1$ and $4x - 3y = 5$.

Correct option: B

Final answer

$y = 1$ and $4x - 3y = 5$

Worked Solutions **Circles**

Q190 MCQ

Find the tangents from (2,6) to $x^2 + y^2 = 20$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-x + 2y = 10$ and $2x + y = 10$.

Correct option: C

Final answer

$-x + 2y = 10$ and $2x + y = 10$

Worked Solutions **Circles**

Q191 **Challenge**

A drone at (2,6) needs tangents to $x^2 + y^2 = 20$. Find both sightline equations and their exact contact points.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$.

Final answer

$-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$

Worked Solutions **Circles**

Q192 Challenge

A drone at (2,6) needs tangents to $x^2 + y^2 = 20$. Find both sightline equations and their exact contact points:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$.

Correct option: B

Final answer

$-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$

Worked Solutions **Circles**

Q193 Written

Classify $x^2 + y^2 = 4$ and $(x-4)^2 + y^2 = 9$.

Solution

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: intersect at 2 points.

Final answer

intersect at 2 points

Q194 Written

Classify $x^2 + y^2 = 9$ and $(x-3)^2 + (y-4)^2 = 4$ from A-E.

Solution

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: B : touch externally.

Final answer

B : touch externally

Worked Solutions **Circles**

Q195 Written

Classify $(x-1)^2+y^2=5^2$ and $x^2+y^2+2x-6y=3$ from A-E.

Solution

1. Find the centre distance D and both radii.
2. Compare D with r_1+r_2 and $|r_1-r_2|$.
3. Relative position: $E\,:\,$ intersect at 2 points.

Final answer

$E\,:\,$ intersect at 2 points

Q196 Written

Classify $x^2+y^2=4$ and $(x-1)^2+y^2=1$ from A-E.

Solution

1. Find the centre distance D and both radii.
2. Compare D with r_1+r_2 and $|r_1-r_2|$.
3. Relative position: $C\,:\,$ touch internally.

Final answer

$C\,:\,$ touch internally

Worked Solutions **Circles**

Q197 Written

Classify $x^2 + y^2 = 18$ and $(x+2)^2 + (y-2)^2 = 2$ from A-E.

Solution

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: $C\,:\,$ touch internally .

Final answer

$C\,:\,$ touch internally

Q198 Written

Classify $(x+3)^2 + y^2 = 16$ and $x^2 + y^2 - 2x - 8y - 8 = 0$ from A-E.

Solution

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: $E\,:\,$ intersect at 2 points .

Final answer

$E\,:\,$ intersect at 2 points

Q199 Written

Classify $x^2 + (y-3)^2 = 45$ and $x^2 + y^2 + 2x - 2y = 3$ from A-E.

Solution

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: D : one is inside the other.

Final answer

D : one is inside the other

Worked Solutions **Circles**

Q200 *MCQ*

For the expression Classify $x^2 + y^2 = 4$ and $(x - 4)^2 + y^2 = 9$., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is intersect at 2 points.

Correct option: C

Final answer

intersect at 2 points

Worked Solutions **Circles**

Q201 MCQ

For the expression Classify $x^2 + y^2 = 9$ and $(x-3)^2 + (y-4)^2 = 4$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $B:$ touch externally.

Correct option: D

Final answer

$B:$ touch externally

Worked Solutions **Circles**

Q202 MCQ

For the expression Classify $(x-1)^2 + y^2 = 5^2$ and $x^2 + y^2 + 2x - 6y = 3$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $E\ :\ \text{intersect at}\,2\,\text{points}$.

Correct option: B

Final answer

$E\ :\ \text{intersect at}\,2\,\text{points}$

Worked Solutions **Circles**

Q203 MCQ

For the expression Classify $x^2 + y^2 = 4$ and $(x-1)^2 + y^2 = 1$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is C : touch internally.

Correct option: A

Final answer

C : touch internally

Worked Solutions **Circles**

Q204 MCQ

For the expression Classify $x^2 + y^2 = 18$ and $(x+2)^2 + (y-2)^2 = 2$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is C : touch internally.

Correct option: D

Final answer

C : touch internally

Worked Solutions **Circles**

Q205 MCQ

For the expression Classify $(x+3)^2+y^2=16$ and $x^2+y^2-2x-8y-8=0$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $E:$ intersect at 2 points.

Correct option: B

Final answer

$E:$ intersect at 2 points

Worked Solutions **Circles**

Q206 MCQ

For the expression Classify $x^2 + (y-3)^2 = 45$ and $x^2 + y^2 + 2x - 2y = 3$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $D:$ one is inside the other.

Correct option: C

Final answer

$D:$ one is inside the other

Q207 Written

Find the circle with centre (3,1) touching $(x-1)^2+(y+3)^2=5$ internally.

Solution

1. Find the centre distance D and both radii.
2. Compare D with r_1+r_2 and $|r_1-r_2|$.
3. Relative position: $(x-3)^2+(y-1)^2=45$.

Final answer

$(x-3)^2+(y-1)^2=45$

Q208 Written

Find the circle with centre (-3,-9) touching $(x+6)^2+(y+5)^2=1$ externally.

Solution

1. Find the centre distance D and both radii.
2. Compare D with r_1+r_2 and $|r_1-r_2|$.
3. Relative position: $(x+3)^2+(y+9)^2=16$.

Final answer

$(x+3)^2+(y+9)^2=16$

Worked Solutions **Circles**

Q209 **Written**

Find the circle with centre (3,-9) touching $(x-3)^2+y^2=9$ internally.

Solution

1. Find the centre distance D and both radii.
2. Compare D with r_1+r_2 and $|r_1-r_2|$.
3. Relative position: $(x-3)^2+(y+9)^2=144$.

Final answer

$(x-3)^2+(y+9)^2=144$

Worked Solutions **Circles**

Q210 MCQ

Find the circle with centre (3,1) touching $(x-1)^2+(y+3)^2=5$ internally:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-3)^2+(y-1)^2=45$.

Correct option: A

Final answer

$(x-3)^2+(y-1)^2=45$

Worked Solutions **Circles**

Q211 MCQ

Find the circle with centre (-3,-9) touching $(x+6)^2+(y+5)^2=1$ externally:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+3)^2+(y+9)^2=16$.

Correct option: D

Final answer

$(x+3)^2+(y+9)^2=16$

Worked Solutions **Circles**

Q212 MCQ

Find the circle with centre (3,-9) touching $(x-3)^2+y^2=9$ internally:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-3)^2+(y+9)^2=144$.

Correct option: A

Final answer

$(x-3)^2+(y+9)^2=144$

Worked Solutions **Circles**

Q213 Challenge

Find the circle with centre (-1,-2) touching $x^2 + y^2 - 4x - 4y + 4 = 0$ externally.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $(x + 1)^2 + (y + 2)^2 = 9$.

Final answer

$(x + 1)^2 + (y + 2)^2 = 9$

Worked Solutions **Circles**

Q214 Challenge

Find the circle with centre (-1,-2) touching $x^2 + y^2 - 4x - 4y + 4 = 0$ externally:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y + 2)^2 = 9$.

Correct option: D

Final answer

$(x + 1)^2 + (y + 2)^2 = 9$

Worked Solutions **Circles**

Q215 Written

Find the line through the intersections of $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x - 4y = 9$.

Solution

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $2x - 4y - 5 = 0$(equivalently $y = x/2 - 5/4$).

Final answer

$2x - 4y - 5 = 0$(equivalently $y = x/2 - 5/4$)

Q216 Written

Find the line through the intersections of $x^2 + y^2 = 6$ and $x^2 + y^2 + 3x + 3y = 9$.

Solution

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $x + y - 1 = 0$.

Final answer

$x + y - 1 = 0$

Worked Solutions **Circles**

Q217 **Written**

Find the line through the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 - 12x + 6y = 3$.

Solution

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $12x - 6y - 7 = 0$.

Final answer

$12x - 6y - 7 = 0$

Q218 **Written**

Find the line through the intersections of $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 4$.

Solution

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $x = 1/2$.

Final answer

$x = 1/2$

Worked Solutions **Circles**

Q219 MCQ

Find the line through the intersections of $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x - 4y = 9$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2x - 4y - 5 = 0$(equivalently $y = x/2 - 5/4$).

Correct option: A

Final answer

$2x - 4y - 5 = 0$(equivalently $y = x/2 - 5/4$)

Worked Solutions **Circles**

Q220 MCQ

Find the line through the intersections of $x^2 + y^2 = 6$ and $x^2 + y^2 + 3x + 3y = 9$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x + y - 1 = 0$.

Correct option: C

Final answer

$x + y - 1 = 0$

Worked Solutions **Circles**

Q221 **MCQ**

Find the line through the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 - 12x + 6y = 3$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $12x - 6y - 7 = 0$.

Correct option: B

Final answer

$12x - 6y - 7 = 0$

Worked Solutions **Circles**

Q222 MCQ

Find the line through the intersections of $x^2 + y^2 = 4$ and $(x-1)^2 + y^2 = 4$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x = 1/2$.

Correct option: D

Final answer

$x = 1/2$

Worked Solutions **Circles**

Q223 **Written**

Find the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 + 2x - y = 5$.

Solution

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $(-3, -1)$ and $(-1, 3)$.

Final answer

$(-3, -1)$ and $(-1, 3)$

Q224 **Written**

Find the intersections of $x^2 + y^2 = 4$ and $(x - 4)^2 + (y - 2)^2 = 16$.

Solution

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $(0, 2)$ and $(8/5, -6/5)$.

Final answer

$(0, 2)$ and $(8/5, -6/5)$

Worked Solutions **Circles**

Q225 Written

Find the intersections of $x^2 + y^2 = 5$ and $(x - 2)^2 + (y - 2)^2 = 1$.

Solution

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: (1, 2) and (2, 1).

Final answer

(1, 2) and (2, 1)

Worked Solutions **Circles**

Q226 *MCQ*

Find the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 + 2x - y = 5$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-3, -1)$ and $(-1, 3)$.

Correct option: B

Final answer

$(-3, -1)$ and $(-1, 3)$

Worked Solutions **Circles**

Q227 MCQ

Find the intersections of $x^2 + y^2 = 4$ and $(x-4)^2 + (y-2)^2 = 16$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(0,\ 2)$ and $(8/5,\ -6/5)$.

Correct option: C

Final answer

$(0,\ 2)$ and $(8/5,\ -6/5)$

Worked Solutions **Circles**

Q228 MCQ

Find the intersections of $x^2 + y^2 = 5$ and $(x-2)^2 + (y-2)^2 = 1$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is (1, 2) and (2, 1).

Correct option: B

Final answer

(1, 2) and (2, 1)

Worked Solutions **Circles**

Q229 **Challenge**

Two radar circles are $x^2 + y^2 = 25$ and $(x + 5)^2 + (y + 5)^2 = 65$. Find their intersection points.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $(-4, 3)$ and $(3, -4)$.

Final answer

$(-4, 3)$ and $(3, -4)$

Worked Solutions **Circles**

Q230 Challenge

Two radar circles are $x^2 + y^2 = 25$ and $(x + 5)^2 + (y + 5)^2 = 65$. Find their intersection points:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-4,\ 3)$ and $(3,\ -4)$.

Correct option: C

Final answer

$(-4,\ 3)$ and $(3,\ -4)$

Worked Solutions **Circles**

Q231 **Written**

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 5 = 0$ and $x^2 + y^2 = 5$, passing through (3,0).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 2x + 6y - 15 = 0$.

Final answer

$x^2 + y^2 + 2x + 6y - 15 = 0$

Q232 **Written**

Find the circle through the intersections of $x^2 + y^2 - 4x - 2y - 8 = 0$ and $x^2 + y^2 = 4$, passing through the origin.

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 4x + 2y = 0$.

Final answer

$x^2 + y^2 + 4x + 2y = 0$

Worked Solutions **Circles**

Q233 **Written**

Find the circle through the intersections of $x^2 + y^2 - 2x - 4y + 3 = 0$ and $x^2 + y^2 = 4$, passing through (3,0).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 10x + 20y - 39 = 0$.

Final answer

$x^2 + y^2 + 10x + 20y - 39 = 0$

Q234 **Written**

Find the circle through the intersections of $x^2 + y^2 - 6x - 8y = 0$ and $x^2 + y^2 - 1 = 0$, passing through (-3,2).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 24x + 32y - 5 = 0$.

Final answer

$x^2 + y^2 + 24x + 32y - 5 = 0$

Worked Solutions **Circles**

Q235 **Written**

Find the circle through the intersections of $x^2 + y^2 - 2x - 2y = 0$ and $x^2 + y^2 - 6x - 4y + 12 = 0$, passing through (4,0).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 - 10x - 6y + 24 = 0$.

Final answer

$x^2 + y^2 - 10x - 6y + 24 = 0$

Worked Solutions **Circles**

Q236 **MCQ**

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 5 = 0$ and $x^2 + y^2 = 5$, passing through (3,0):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 2x + 6y - 15 = 0$.

Correct option: D

Final answer

$x^2 + y^2 + 2x + 6y - 15 = 0$

Worked Solutions **Circles**

Q237 MCQ

Find the circle through the intersections of $x^2 + y^2 - 4x - 2y - 8 = 0$ and $x^2 + y^2 = 4$, passing through the origin:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 4x + 2y = 0$.

Correct option: A

Final answer

$x^2 + y^2 + 4x + 2y = 0$

Worked Solutions **Circles**

Q238 MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 4y + 3 = 0$ and $x^2 + y^2 = 4$, passing through (3,0):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 10x + 20y - 39 = 0$.

Correct option: D

Final answer

$x^2 + y^2 + 10x + 20y - 39 = 0$

Worked Solutions **Circles**

Q239 MCQ

Find the circle through the intersections of $x^2 + y^2 - 6x - 8y = 0$ and $x^2 + y^2 - 1 = 0$, passing through (-3,2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 24x + 32y - 5 = 0$.

Correct option: C

Final answer

$x^2 + y^2 + 24x + 32y - 5 = 0$

Worked Solutions **Circles**

Q240 **MCQ**

Find the circle through the intersections of $x^2 + y^2 - 2x - 2y = 0$ and $x^2 + y^2 - 6x - 4y + 12 = 0$, passing through (4,0):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 10x - 6y + 24 = 0$.

Correct option: B

Final answer

$x^2 + y^2 - 10x - 6y + 24 = 0$

Worked Solutions **Circles**

Q241 **Challenge**

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 4x - 2y - 11 = 0$, passing through (2,2).

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$.

Final answer

$x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$

Worked Solutions **Circles**

Q242 **Challenge**

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 4x - 2y - 11 = 0$, passing through (2,2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$.

Correct option: A

Final answer

$x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$

Worked Solutions **Circles**

Quiz MCQ 1 **Concept Quiz · MCQ**

For the circle $(x-2)^2+(y+3)^2=25$, determine its centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $\text{centre } (2,\,-3),\ \text{radius } 5$.

Correct option: B

Final answer

$\text{centre } (2,\,-3),\ \text{radius } 5$

Worked Solutions **Circles**

Quiz MCQ 2 Concept Quiz · MCQ

Determine the substitution discriminant when a line meets a circle at exactly one real point:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is 0.

Correct option: D

Final answer

0

Worked Solutions **Circles**

Quiz MCQ 3 Concept Quiz · MCQ

Determine the chord length for radius r and centre-to-chord distance d:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2\sqrt{r^2 - d^2}$.

Correct option: A

Final answer

$2\sqrt{r^2 - d^2}$

Worked Solutions **Circles**

Quiz MCQ 4 Concept Quiz · MCQ

Determine the relative position of two non-overlapping circles when $D = r1 + r2$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is externally tangent.

Correct option: D

Final answer

externally tangent

Worked Solutions **Circles**

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the circle with centre (1,-2) passing through (4,2).

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2+(y-k)^2=r^2$ and compare coefficients when needed.
3. Circle: $(x-1)^2+(y+2)^2=25$.

Final answer

$(x-1)^2+(y+2)^2=25$

Quiz WRITTEN 6 **Concept Quiz · Written**

Find the intersections of $x^2+y^2=25$ and $y=3$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(4,\ 3)\,,\ (-4,\ 3)$.

Final answer

$(4,\ 3)\,,\ (-4,\ 3)$

Worked Solutions **Circles**

Quiz WRITTEN 7 **Concept Quiz · Written**

Find the tangent to $x^2 + y^2 = 25$ at (3,4).

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x-h)(x_1-h)+(y-k)(y_1-k)=r^2$ and rearrange.
3. Tangent: $3x + 4y = 25$.

Final answer

$3x + 4y = 25$

Quiz WRITTEN 8 **Concept Quiz · Written**

The circles $S_1 = x^2 + y^2 - 5 = 0$ and $S_2 = x^2 + y^2 - 2x - 2y - 4 = 0$ meet. Find the circle $S_1 + \lambda S_2 = 0$ passing through (1,1).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $\lambda = -1/2; x^2 + y^2 + 2x + 2y - 6 = 0$.

Final answer

$\lambda = -1/2; x^2 + y^2 + 2x + 2y - 6 = 0$

Answer key

Use the question number shown on each page.

Q001 $(x-3)^2+(y+2)^2=25$

Q002 $(x+3)^2+(y-1)^2=4$

Q003 $x^2+y^2=16$

Q004 $(x-3)^2+(y-4)^2=36$

Q005 $(x+2)^2+(y-6)^2=12$

Q006 $x^2+y^2=25$

Q007 A

Q008 A

Q009 D

Q010 B

Q011 B

Q012 A

Q013 circle; centre $(3, -1)$, radius 2

Q014 circle; centre $(2, 3)$, radius 3

Q015 circle; centre $(2, -4)$, radius 4

Q016 circle; centre $(2, 0)$, radius 2

Q017 circle; centre $(-1, 2)$, radius 5

Q018 circle; centre $(-2, 3)$, radius $\sqrt{13}$

Q019 A

Q020 C

Q021 D

Q022 C

Q023 C

Q024 C

Q025 centre $(5, 3)$, radius$^2=13$; $(x-5)^2+(y-3)^2=13$

Q026 B

Q027 $(x-2)^2+(y-3)^2=13$

Q028 $(x+3)^2+(y-1)^2=58$

Q029 $(x-1)^2+(y+3)^2=25$

Q030 $(x+4)^2+(y-1)^2=58$

Q031 $(x-2)^2+(y+1)^2=5$

Q032 $x^2+y^2=25$

Q033 C

Q034 A

Q035 D

Q036 A

Q037 B

Q038 A

Q039 $(x-2)^2+(y+1)^2=17$

Q040 $(x-4)^2+(y-1)^2=10$

Q041 $(x-1)^2+(y-3)^2=25$

Q042 $(x+1)^2+(y+2)^2=25$

Q043 $(x+2)^2+(y-1)^2=2$

Q044 $x^2+(y-2)^2=25$

Q045 A

Q046 C

Q047 C

Q048 A

Q049 B

Q050 C

Q051 $x^2+y^2-5x-y+4=0$

Q052 $x^2+y^2-3x-7y+6=0$

Q053 $x^2+y^2-2x-4y=0$

Q054 $x^2+y^2-6y-16=0$

Q055 $x^2+y^2-2x+8y-8=0$

Q056 A

Q057 B

Q058 D

Q059 D

Q060 D

Q061 $x^2+y^2-6x-6y+8=0$; centre $(3, 3)$, radius $\sqrt{10}$

Q062 A

Q063 $(x-4)^2+(y+2)^2=4$

Q064 $(x+2)^2+(y-3)^2=4$

Q065 $(x+1)^2+(y-4)^2=16$

Q066 $(x-3)^2+(y+4)^2=16$

Q067 $(x-5)^2+(y+4)^2=16$

Q068 $(x+1)^2+(y-6)^2=1$

Q069 $(x-2)^2+(y+5)^2=4$

Q070 $(x+3)^2+(y+4)^2=9$

Q071 B

Q072 B

Q073 B

Q074 D

Q075 D

Q076 D

Q077 B

Q078 C

Q079 $(x+1)^2+(y-1)^2=1$ or $(x+5)^2+(y-5)^2=25$

Q080 $(x-1)^2+(y-1)^2=1$ or $(x-5)^2+(y-5)^2=25$

Q081 $(x+1)^2+(y+1)^2=1$ or $(x+5)^2+(y+5)^2=25$

Q082 $(x+1)^2+(y+1)^2=1$ or $(x+5)^2+(y+5)^2=25$

Q083 $(x-5)^2+(y+5)^2=25$ or $(x-13)^2+(y+13)^2=169$

Q084 C

Q085 D

Q086 C

Q087 C

Q088 B

Q089 $(x-1)^2+(y+1)^2=25$

Q090 $(x+5/2)^2+(y-5/2)^2=25/2$

Q091 $(x-2)^2+(y-3)^2=2$

Q092 $(x-4)^2+(y-2)^2=26$

Q093 D

Q094 C

Q095 C

Q096 D

Q097 $(x+1)^2+y^2=5$; centre $(-1, 0)$, radius $\sqrt{5}$

Q098 C

Q099 $k<-1$ or $k>3$

Q100 $k<-\sqrt{6}$ or $k>\sqrt{6}$

Q101 $k<5/4$

Q102 $k<-1/5$ or $k>3$

Q103 $-1<k<4$

Q104 C

Q105 B

Q106 C

Q107 B

Q108 D

Q109 $0 < k < 4$

Q110 B

Q111 $(1, -1)$ and $(-1, 3)$

Q112 $(0, 2)$ and $(2, 0)$

Q113 $(3, -1)$

Q114 $(-2, 1)$

Q115 $(-2, 1)$ and $(4, -1)$

Q116 C

Q117 D

Q118 C

Q119 D

Q120 B

Q121 0 points

Q122 2 points : $(1, -3)$ and $(3, 1)$

Q123 2 points : $\left(-1+\sqrt{6}/2,\ 1+\sqrt{6}/2\right)$ and $\left(-1-\sqrt{6}/2,\ 1-\sqrt{6}/2\right)$

Q124 0 points

Q125 1 point : $(-2, 0)$

Q126 D

Q127 A

Q128 C

Q129 C

Q130 D

Q131 $-5 < k < 5$

Q132 $r = \sqrt{5}$

Q133 $m = \pm\sqrt{3}$

Q134 $0 < r < \sqrt{10}$

Q135 $m = \pm 4$

Q136 $n < -10$ or $n > 10$

Q137 $-5 <= m <= 5$

Q138 $r = \sqrt{2}$

Q139 $r > \sqrt{17}$

Q140 D

Q141 B

Q142 B

Q143 B

Q144 C

Q145 A

Q146 C

Q147 B

Q148 D

Q149 $0 < r < 2$

Q150 D

Q151 $\sqrt{14}$

Q152 4

Q153 $2\sqrt{5}$

Q154 $3\sqrt{2}$

Q155 2

Q156 A

Q157 C

Q158 D

Q159 B

Q160 D

Q161 $k = \pm 2$

Q162 $k = \pm\sqrt{31}$

Q163 $k = \pm 2$

Q164 $k = 0$ or $k = -2$

Q165 C

Q166 D

Q167 A

Q168 B

Q169 $k = -1$ or $k = 7$

Q170 A

Q171 $4x + 3y = 25$

Q172 $5x + 3y = 34$

Q173 $3x - y = 10$

Q174 $-5x - 2y = 29$

Q175 $-2x + y = 5$

Q176 A

Q177 C

Q178 A

Q179 C

Q180 A

Q181 $2x + y = 5$ and $-x + 2y = 5$

Q182 $-3x + y = 10$ and $x + 3y = 10$

Q183 $-2x + y = 5$ and $x + 2y = 5$

Q184 $y = 1$ and $4x - 3y = 5$

Q185 $-x + 2y = 10$ and $2x + y = 10$

Q186 D

Q187 A

Q188 D

Q189 B

Q190 C

Q191 $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$

Q192 B

Q193 intersect at 2 points

Q194 B : touch externally

Q195 E : intersect at 2 points

Q196 C : touch internally

Q197 C : touch internally

Q198 E : intersect at 2 points

Q199 D : one is inside the other

Q200 C

Q201 D

Q202 B

Q203 A

Q204 D

Q205 B

Q206 C

Q207 $(x-3)^2 + (y-1)^2 = 45$

Q208 $(x+3)^2 + (y+9)^2 = 16$

Q209 $(x-3)^2 + (y+9)^2 = 144$

Q210 A

Q211 D

Q212 A

Q213 $(x+1)^2 + (y+2)^2 = 9$

Q214 D

Q215 $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4$)

Q216 $x + y - 1 = 0$

Q217 $12x - 6y - 7 = 0$

Q218 $x = 1/2$

Q219 A

Q220 C

Q221 B

Q222 D

Q223 $(-3, -1)$ and $(-1, 3)$

Q224 $(0, 2)$ and $(8/5, -6/5)$

Q225 $(1, 2)$ and $(2, 1)$

Q226 B

Q227 C

Q228 B

Q229 $(-4, 3)$ and $(3, -4)$

Q230 C

Q231 $x^2 + y^2 + 2x + 6y - 15 = 0$

Q232 $x^2 + y^2 + 4x + 2y = 0$

Q233 $x^2 + y^2 + 10x + 20y - 39 = 0$

Q234 $x^2 + y^2 + 24x + 32y - 5 = 0$

Q235 $x^2 + y^2 - 10x - 6y + 24 = 0$

Q236 D

Q237 A

Q238 D

Q239 C

Q240 B

Q241 $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$

Q242 A

Quiz MCQ 1 B

Quiz MCQ 2 D

Quiz MCQ 3 A

Quiz MCQ 4 D

Quiz WRITTEN 5 $(x-1)^2 + (y+2)^2 = 25$

Quiz WRITTEN 6 $(4, 3), (-4, 3)$

Quiz WRITTEN 7 $3x + 4y = 25$

Quiz WRITTEN 8 $\lambda = -1/2; x^2 + y^2 + 2x + 2y - 6 = 0$